Finals Sketchbook

35 classes · All Subjects · Fall 2026 finals

Each class: the idea in one line, the sketch step by step, plain explanation, formulas, a worked example, common traps and where to read more.

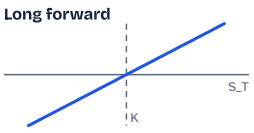
Contents

1. The four derivatives and their payoffs (Derivatives)
2. How often interest is added (Derivatives)
3. Forward price by no-arbitrage (Derivatives)
4. Futures, margin and daily settlement (Derivatives)
5. Hedging with futures (Derivatives)
6. Forward rates, duration and convexity (Derivatives)
7. Swaps: trading payment streams (Derivatives)
8. Option strategies and put-call parity (Derivatives)
9. Binomial trees for options (Derivatives)
10. Black-Scholes and the Greeks (Derivatives)
11. Money today vs money later (Valuation)
12. Loans: payments, default and prepayment (Valuation)
13. Bonds and building a zero curve (Valuation)
14. Capital budgeting: NPV and IRR (Valuation)
15. Simulation, correlation and Cholesky (Valuation)
16. Random walks and stock price paths (Valuation)
17. Portfolio theory and the CAPM (Valuation)
18. ETFs and futures for end users (Valuation)
19. CDO waterfalls and tranches (Valuation)
20. Volatility smiles and implied densities (Valuation)
21. Real options and the Merton model (Valuation)
22. Volatility, diversification, VaR and ES (Risk Management)
23. Immunization and asset-liability management (Risk Management)
24. Currency and commodity risk (Risk Management)
25. Greeks and delta hedging (Risk Management)
26. Credit risk: default, loss and spreads (Risk Management)
27. Counterparty risk and CVA (Risk Management)
28. Operational risk and controls (Risk Management)
29. Bank capital and liquidity rules (Risk Management)
30. P vs Q, risk drivers and stress tests (Risk Management)
31. Bloomberg: passing the four tests (Bloomberg)
32. Economic indicators (Bloomberg)
33. Currencies (Bloomberg)
34. Fixed income and the yield curve (Bloomberg)
35. Equities and valuation multiples (Bloomberg)

The four derivatives and their payoffs

A derivative is a contract whose value comes from something else (a stock, a rate, oil). There are four basic kinds: forwards, futures, swaps and options.

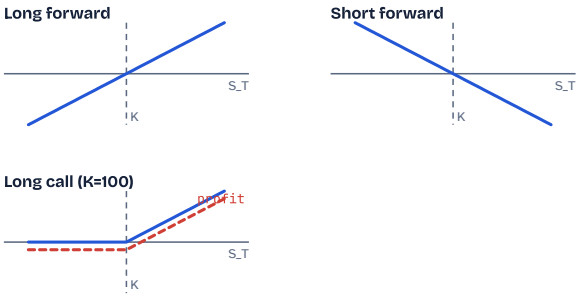
The sketch, step by step



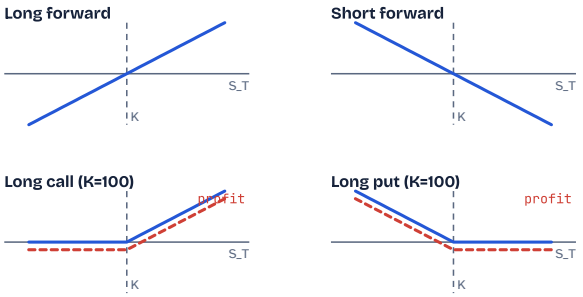
Step 1. Long forward: you win $1 for every $1 the price ends above K, and lose the same below.



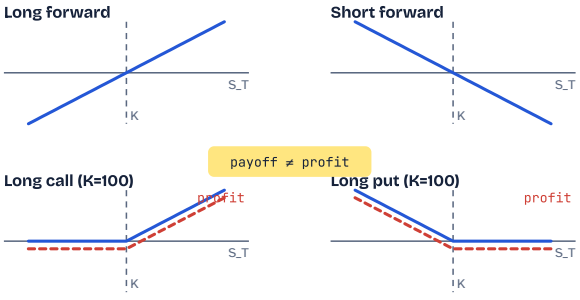
Step 2. Short forward: the mirror image. Your gain is exactly the other side's loss.



Step 3. Long call: you only use it if the price is above K. Dashed line = profit after paying the premium.



Step 4. Long put: you profit when the price falls below K. Insurance for a stock owner.



Step 5. Four shapes to remember: two straight lines, two "hockey sticks".

What it means

- **Forward:** agree today to buy (long) or sell (short) at a fixed price K on a later date. Both sides must go through with it. Made privately (over the counter).
- **Future:** the same idea, but traded on an exchange, standardized, and settled every day through margin (Class 7).
- **Swap:** trade one stream of payments for another, e.g. fixed interest for floating (Class 9).
- **Option:** the *right but not the obligation*. A call is the right to buy at K, a put the right to sell at K. You pay a premium up front for that right.

Payoff is what you get at the end. **Profit** is payoff minus what you paid. Draw the picture every time; it stops sign mistakes.

The formulas

```
Long forward payoff  = S_T - K
Short forward payoff = K - S_T
Long call payoff     = max(S_T - K, 0)
Long put payoff      = max(K - S_T, 0)
Profit = payoff - premium paid
```

S_T = price of the asset at the end, K = agreed (strike/delivery) price.

Worked example

Question. You buy a call with strike $100 for a $5 premium. At expiry the stock is $120. What is your payoff and profit? What if the stock is $90?

1. At $120: payoff = max(120 − 100, 0) = `20`
2. Profit = 20 − 5 = `15`
3. At $90: payoff = max(90 − 100, 0) = `0` (you just do not use the option)
4. Profit = 0 − 5 = `-5`. The most a call buyer can lose is the premium.

Answer: At $120: payoff 20, profit 15. At $90: payoff 0, profit −5

Common traps

- Mixing payoff with profit.
- Saying a forward costs money to enter. A fair forward costs nothing up front.
- Forgetting the short side of an option can lose a lot.

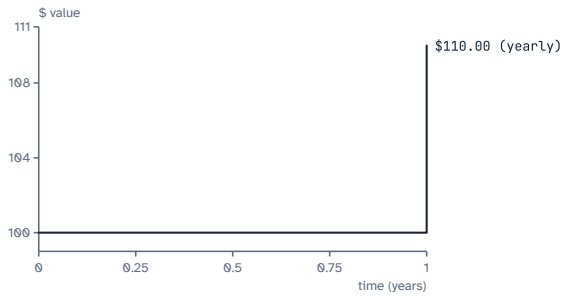
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_02_Lecture_01_Introduction_to_Derivatives.pdf`
- `Hull 8e, Chapters 1 and 10`

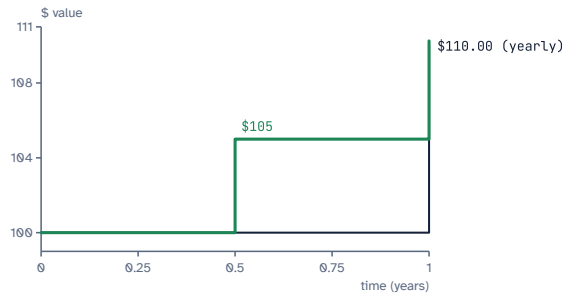
How often interest is added

"10% a year" can mean different amounts of money depending on how often interest is added. Continuous compounding is the limit where it is added every instant.

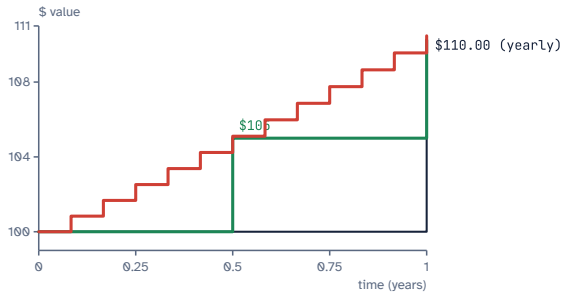
The sketch, step by step



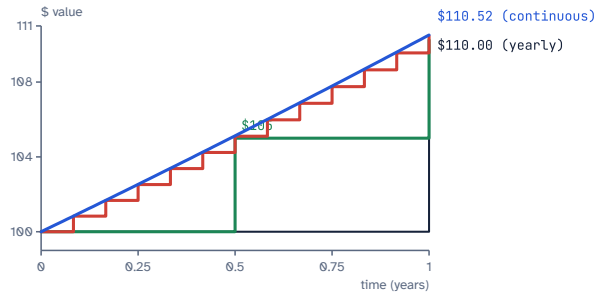
Step 1. $100 at 10% a year. Added once a year, it jumps to $110 at the end.



Step 2. Added twice a year: $105 after 6 months, then 5% on $105 gives $110.25.



Step 3. Monthly: 12 tiny jumps. You end at $110.47.



Step 4. Continuous: a smooth curve, interest every instant. You end at $110.52.



Step 5. The more often you compound, the more you get, but it levels off at e^0.10.

What it means

If the bank adds interest twice a year, you earn interest on your interest sooner, so you end up with a bit more. Same headline rate, different result. That is why every rate needs a **compounding frequency** attached.

- **m times a year:** grow by (1 + R/m) each piece, m pieces a year.
- **Continuous:** grow by e^{Rt}. Derivatives math (forwards, Black-Scholes) almost always uses this.
- **Converting:** two rates are "the same" if they grow $1 to the same amount in one year.

The formulas

```
Compounded m times:  FV = A × (1 + R/m)^(m·t)
Continuous:          FV = A × e^(R·t)
Convert m → cont:    Rc = m × ln(1 + Rm/m)
Convert cont → m:    Rm = m × (e^(Rc/m) - 1)
```

A = amount, R = annual rate, m = times per year, t = years, ln = natural log.

Worked example

Question. You have 10% a year compounded twice a year. What is the continuously compounded rate?

1. Each half year you grow by 1 + 0.10/2 = `1.05`
2. Use the convert rule: `Rc = 2 × ln(1.05)`
3. ln(1.05) = 0.04879, times 2 = `0.09758`
4. Check: e^0.09758 = 1.1025, and 1.05^2 = 1.1025. Same growth.

Answer: Rc = 9.758% continuous

Calculator example

Inputs: Annual rate (%) = 10, Times per year = 2

```
$100 after 1 year = 100 × (1 + 0.1/2)^2 = $110.25
Same as continuous rate Rc = 2 × ln(1 + 0.0500) = 9.758%
```

Common traps

- Plugging a semiannual rate into e^(rT) without converting it.
- Using t in months when the rate is per year.
- Thinking a higher m changes the headline rate. It only changes how much you actually earn.

Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx`
- `Hull 8e, Chapter 4 (Interest Rates)`

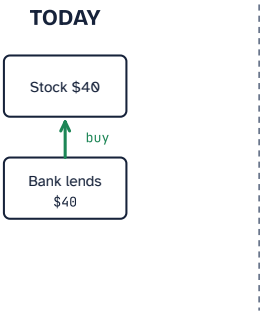
Forward price by no-arbitrage

The fair forward price is what it costs to buy the asset now with borrowed money and carry it to the delivery date. Any other price gives someone free money.

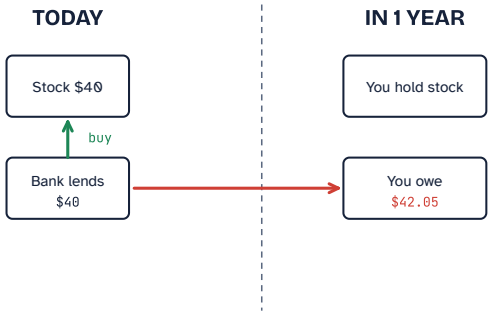
The sketch, step by step



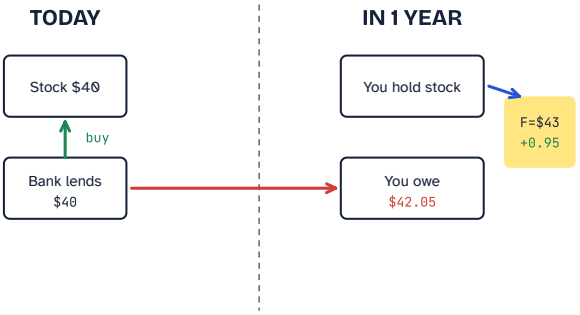
Step 1. Today the stock costs $40. You can borrow at 5%.



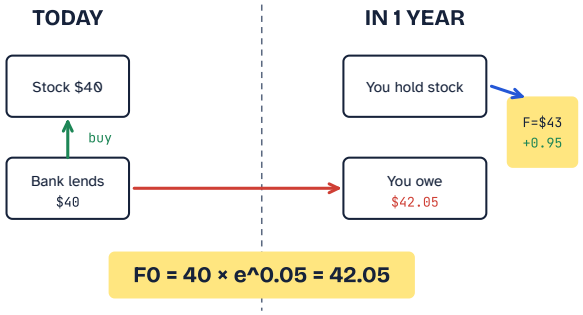
Step 2. Strategy: borrow $40 and buy the stock. Costs you nothing today.



Step 3. In one year you own the stock and owe 40 × e^0.05 = $42.05.



Step 4. If someone offers a forward at $43, sell it to them. Deliver the stock, get 43, repay 42.05. Free $0.95.



Step 5. So the only price with no free money is F0 = $42.05.

What it means

Arbitrage means a riskless profit with no money down. Markets hunt it away. So: if you can borrow $40 at 5%, buy the stock, and hold it a year, it costs you $40 × e^0.05 = $42.05 to own the stock in a year. A forward to buy it in a year must also cost $42.05, otherwise you would do one and the opposite of the other.

- **Income** (dividends) makes holding the stock cheaper, so subtract its PV: $F = (S - I)e^{rT}$.
- **Dividend yield q** or a foreign interest rate r_f works the same: $F = S\,e^{(r-q)T}$.

- **Forward price vs forward value:** a new forward is worth 0. An old one with delivery price K is worth $(F_0 - K)e^{-rT}$ to the long side.

The formulas

```
No income:        F0 = S0 × e^(rT)
Known income I:   F0 = (S0 - I) × e^(rT)
Yield q / FX:     F0 = S0 × e^((r - q)T)     (q = r_f for currency)
Value of long:    f  = (F0 - K) × e^(-rT)
```

S0 = spot price now, r = risk-free rate (continuous), T = years, K = delivery price written in an existing contract.

Worked example

Question. A stock is $40, pays no dividend, r = 5%, T = 1 year. (a) Fair forward price? (b) The market quotes $43. How do you make money? (c) An old contract has K = $40. What is it worth?

1. (a) F0 = 40 × e^0.05 = `42.05`
2. (b) $43 is too high. Sell the forward at 43, borrow $40, buy the stock.
3. In a year: deliver the stock, receive 43, repay the loan 42.05. Profit `0.95` with no risk.
4. (c) f = (42.05 − 40) × e^(−0.05) = `1.95` for the long side.

Answer: F0 = 42.05; arbitrage profit 0.95; old contract worth 1.95

Calculator example

Inputs: Spot price S0 = 40, Rate r (%) = 5, Yield q or r_f (%) = 0, Years T = 1, Old contract K = 40

```
Fair forward F0 = 40 × e^(0.0500 × 1) = 42.0508
Value of old long contract = (42.0508 - 40) × e^(-0.05 × 1) = 1.9508
```

Common traps

- Mixing price (the K that makes value 0) and value (what an existing contract is worth).
- Adding dividends instead of subtracting them.
- Doing the arbitrage in the wrong direction. Forward too high means sell the forward and buy the asset.

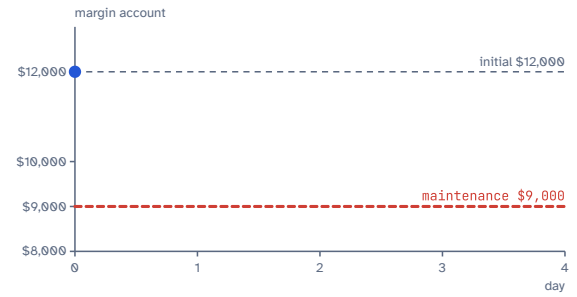
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx`
- `Derivatives/04 Assignments and Solutions/FRE6073_SOLUTION_03_Assignment_01_Solutions.pdf`

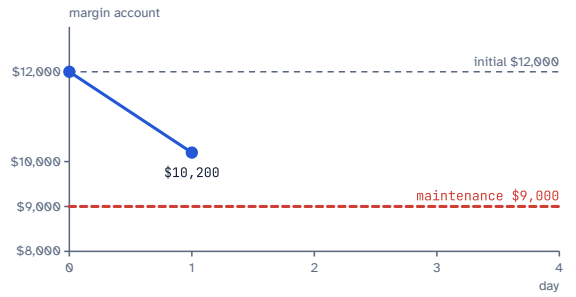
Futures, margin and daily settlement

With futures, gains and losses are paid every day through a margin account, so nobody can build up a big unpaid loss.

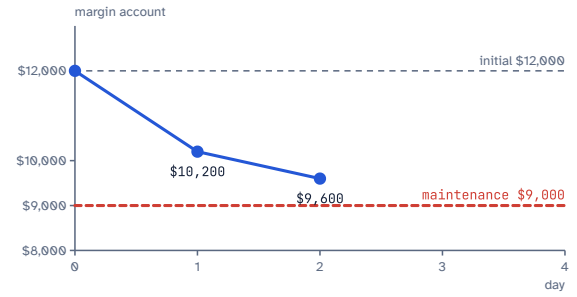
The sketch, step by step



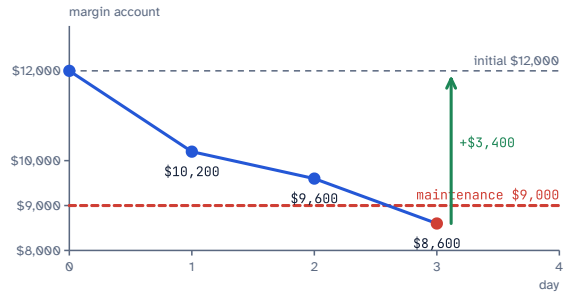
Step 1. You open with $12,000. The dashed red line is the $9,000 maintenance floor.



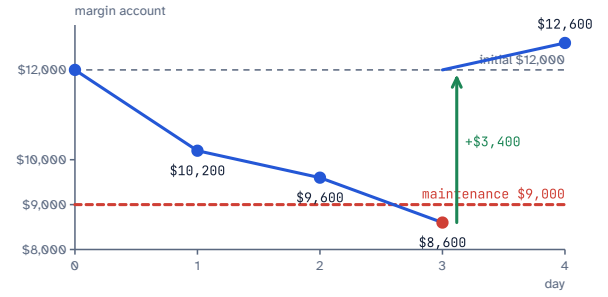
Step 2. Day 1: gold falls $9. You lose $1,800 that day. Balance $10,200.



Step 3. Day 2: down $3 more. Balance $9,600.



Step 4. Day 3: down $5. Balance $8,600 is under the floor. Margin call! Add $3,400 to get back to $12,000.



Step 5. Day 4: gold rises $3. You gain $600. Balance $12,600.

What it means

- **Initial margin:** the deposit to open the position.
- **Marking to market:** each day the exchange moves money between buyers and sellers based on the new futures price.
- **Maintenance margin:** a floor. If your account drops below it you get a **margin call** and must top it back up to the *initial* margin.
- **Clearing house:** stands between buyer and seller, so you never depend on the other trader paying. That removes most counterparty risk.
- **Basis** = spot price − futures price. It goes to zero at delivery.

The formulas

```
Daily gain (long) = (F_today - F_yesterday) × contract size × number of contracts
If balance < maintenance → deposit (initial - balance)
```

Short positions get the opposite sign.

Worked example

Question. You go long 2 gold futures (100 oz each) at $1,250. Initial margin $6,000 per contract, maintenance $4,500. Prices: 1,241 → 1,238 → 1,233 → 1,236. Track the account.

1. Start: 2 × 6,000 = `$12,000`. Each $1 move = 2 × 100 = $200.
2. Day 1: −9 × 200 = −1,800 → `$10,200`
3. Day 2: −3 × 200 = −600 → `$9,600`
4. Day 3: −5 × 200 = −1,000 → `$8,600`, below the $9,000 floor. Margin call: deposit `$3,400` to get back to $12,000.
5. Day 4: +3 × 200 = +600 → `$12,600`

Answer: Margin call of $3,400 on day 3; balance $12,600 after day 4

Common traps

- Topping up only to the maintenance level. You go back to the initial margin.
- Forgetting to multiply by contract size and number of contracts.
- Thinking margin is a fee. It is your money, held as a deposit.

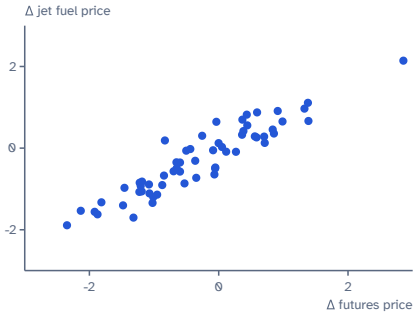
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx`
- `Hull 8e, Chapter 2`

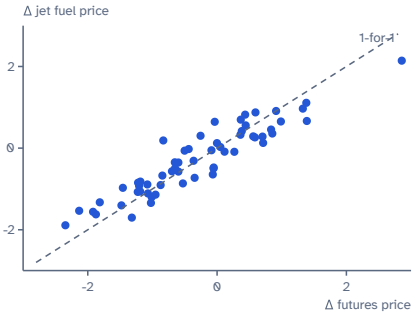
Hedging with futures

A hedge uses futures to cancel out a price risk you already have. The hedge ratio tells you how many futures to use when the futures and your exposure do not move one-for-one.

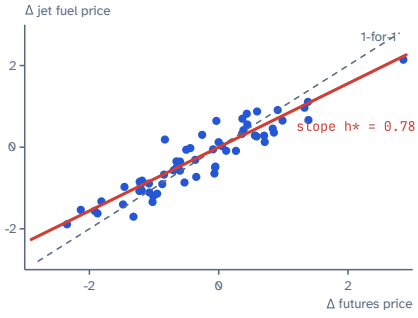
The sketch, step by step



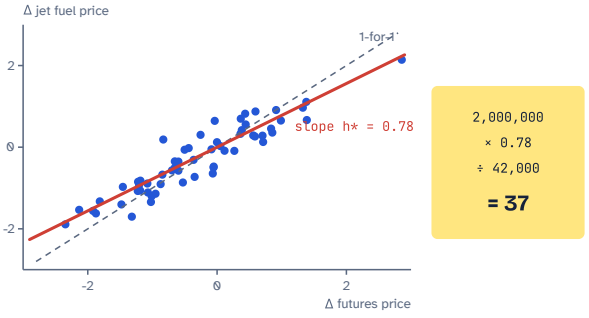
Step 1. Each dot is one month: how much jet fuel moved (up) vs how much heating oil futures moved (across).



Step 2. They move together, but not exactly one-for-one.



Step 3. The best-fit line has slope 0.78. That slope is the hedge ratio h*.



Step 4. So hedge 78% of the 2m gallons: 1.56m gallons ÷ 42,000 = 37 contracts.

What it means

- **Short hedge:** you will sell something later (a farmer's wheat). Sell futures now so a price fall is offset by a futures gain.
- **Long hedge:** you will buy something later (an airline's fuel). Buy futures now.
- **Basis risk:** basis = spot − futures. If you hedge with a different but related product (heating oil for jet fuel), or close out early, the basis can move and the hedge is not perfect.
- **Minimum-variance hedge ratio h*:** the slope when you regress changes in your price on changes in the futures price. It is $\rho \times \sigma_S/\sigma_F$.
- **Number of contracts:** h* × (size of your exposure) / (size of one contract).
- **Stock index hedge:** to remove market risk from a portfolio with beta β, short β × (portfolio value)/(futures price × multiplier) contracts.

The formulas

```
h* = ρ × σS / σF
N* = h* × QA / QF
Index hedge: N* = β × VA / VF
```

σS, σF = std dev of price changes (spot, futures), ρ = their correlation, QA = units exposed, QF = units per contract, VA = portfolio value, VF = futures price × multiplier.

Worked example

Question. An airline will buy 2 million gallons of jet fuel. It hedges with heating oil futures (42,000 gallons each). σS = 0.0263, σF = 0.0313, ρ = 0.928. How many contracts?

1. h* = 0.928 × 0.0263 / 0.0313 = `0.7798`
2. Exposure to hedge = 0.7798 × 2,000,000 = `1,559,520 gallons`
3. Contracts = 1,559,520 / 42,000 = `37.13`
4. Round: buy `37` heating oil futures (a long hedge, since it will buy fuel).

Answer: h* ≈ 0.78, buy about 37 contracts

Calculator example

Inputs: Correlation ρ = 0.928, σ spot = 0.0263, σ futures = 0.0313, Units exposed = 2000000, Units per contract = 42000

```
h* = 0.928 × 0.0263 / 0.0313 = 0.7798
Contracts = 0.7798 × 2,000,000 / 42,000 = 37.13
```

Common traps

- Using h = 1 when the futures is on a different product.
- Going short when you need to buy later (it is a long hedge).
- Forgetting that a hedge also removes your upside.

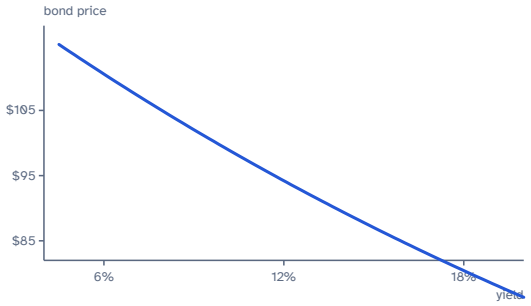
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx`
- `Hull 8e, Chapter 3 (Hedging strategies using futures)`

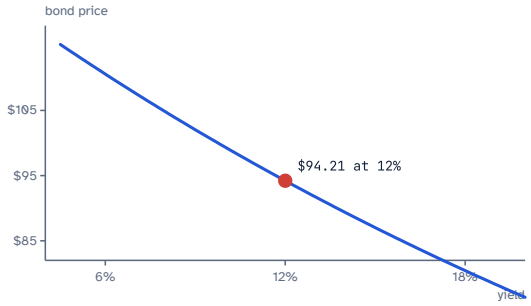
Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

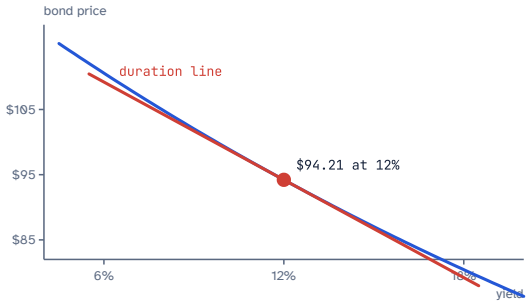
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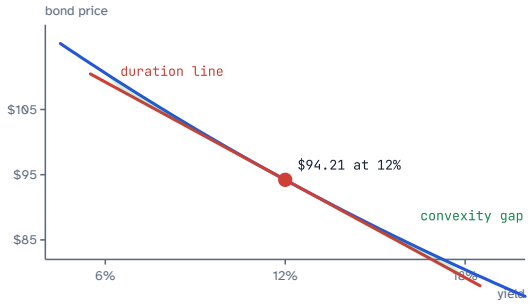
Step 1. The price of the bond at each yield. It slopes down and bends (is "convex").



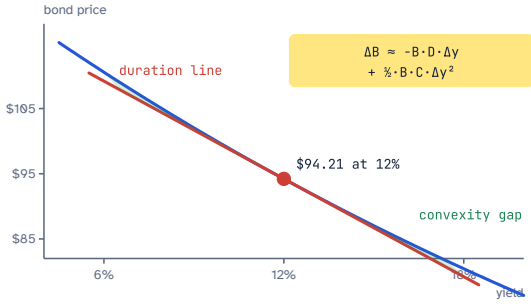
Step 2. At today's yield of 12%, the price is $94.21.



Step 3. Duration draws the tangent line at that point. Slope = −B × D.



Step 4. The tangent is below the curve on both sides. That gap is convexity.



Step 5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

What it means

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always *underestimates* the price after a big move either way. Full repricing is the true answer; duration

is the quick estimate.

The formulas

```
Forward rate:   F(T1,T2) = (R2·T2 - R1·T1) / (T2 - T1)
Duration:       D = Σ t × [CF(t)·e^(-y·t)] / B
Price change:   ΔB ≈ -B × D × Δy  (+ ½ × B × C × Δy²)
```

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Worked example

Question. 3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.
2. Discount each at e^(−0.12t) and add: `B = 94.213`
3. Weight each time by its share of the price and add: `D = 2.653 years`
4. ΔB ≈ −94.213 × 2.653 × 0.001 = `-0.250`, so the price falls to about 93.963.

Answer: B = 94.21, D = 2.653, price drops about $0.25

Common traps

- Using Δy = 0.1 instead of 0.001.
- Forgetting the minus sign: rates up means price down.
- Using duration for a large rate shock without the convexity term.

Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`
- `Risk syllabus RMN2 (duration, convexity, term structure)`

Swaps: trading payment streams

In an interest rate swap, one side pays fixed and receives floating. You value it as owning one bond and owing another.

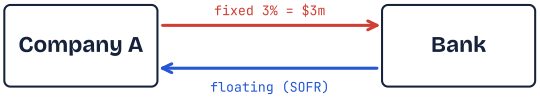
The sketch, step by step



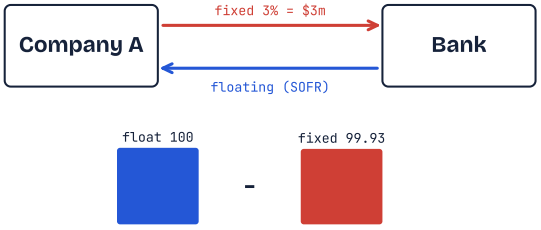
Step 1. Two parties: Company A and a bank.



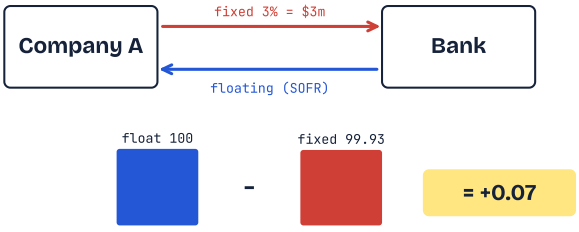
Step 2. A pays the bank a fixed 3% of $100m every year: $3m.



Step 3. The bank pays A the floating rate (e.g. SOFR) on $100m.



Step 4. Same as A owning a floating bond (worth 100) and owing a fixed bond (worth 99.93).



Step 5. Swap value to A = 100 − 99.93 = +0.07 ($m).

What it means

Company A has a floating-rate loan but wants a fixed payment. It enters a swap: pays a fixed 3% to the bank, receives the floating rate back. Only interest moves; the notional (e.g. $100m) never changes hands in an interest rate swap.

- **Bond view:** receive floating + pay fixed = long a floating bond, short a fixed bond. Value = $B_{float} - B_{fixed}$.
- Right after a reset, a floating bond is worth its par value (100).
- **Forward view:** treat each future exchange as a forward rate agreement and add them up. Both views give the same answer.

- **Currency swap:** swap interest *and* principal in two currencies. Value each leg in its own currency, convert at today's spot rate.

The formulas

```
Pay fixed, receive floating:   V = B_float - B_fixed
B_fixed = Σ c·e^(-r_i·t_i) + L·e^(-r_n·t_n)
B_float = L  (just after a reset)
```

L = notional, c = fixed coupon payment, r_i = zero rate for time t_i.

Worked example

Question. $100m swap, 2 years left, you pay 3% fixed annually and receive floating (just reset). Zero rates: 1-yr 2.5%, 2-yr 3.0% (continuous). Value to you?

1. Fixed bond: 3 × e^(−0.025) + 103 × e^(−0.06)
2. = 2.926 + 97.001 = `99.928` ($m)
3. Floating bond right after reset = `100`
4. Value = 100 − 99.928 = `+0.072` ($m). Positive: the 3% you pay is a little below the fair swap rate for this curve (about 3.04%), so the swap is slightly in your favour.

Answer: Swap worth about +$72,000 to the fixed payer

Common traps

- Getting the sign backwards. Write which side you are on first.
- Forgetting the notional at the end of the fixed bond (103, not 3).
- For currency swaps, converting with forward rates in the bond view. Use spot.

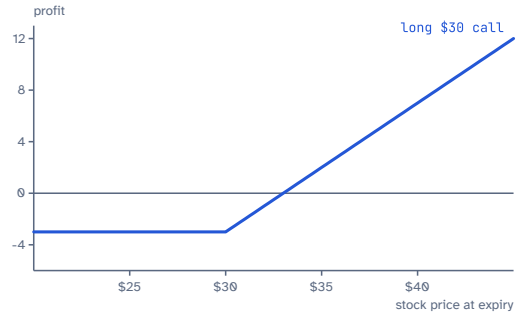
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`
- `Derivatives/04 Assignments and Solutions/FRE6073_ASSIGNMENT_05_Assignment_03.pdf`

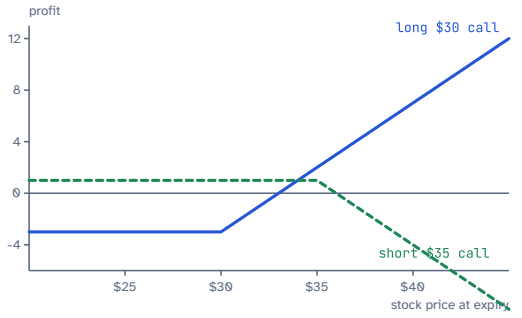
Option strategies and put-call parity

Options can be stacked like Lego to make any payoff shape you want. Put-call parity is the rule that links a call and a put with the same strike.

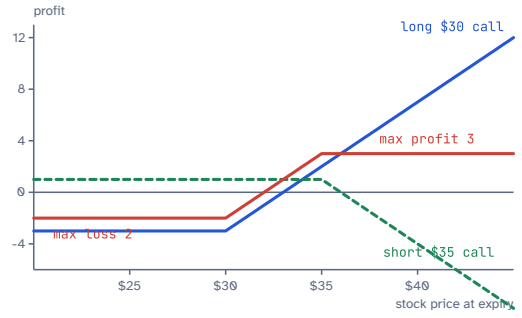
The sketch, step by step



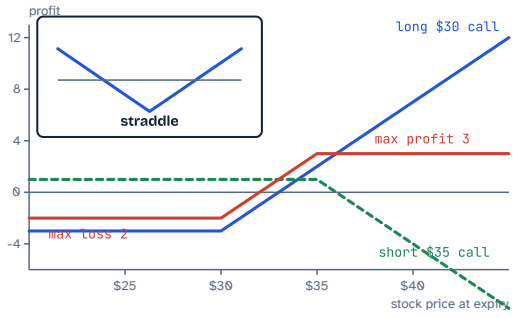
Step 1. Start: buy a $30 call. Hockey stick that starts at 30.



Step 2. Add: sell a $35 call. That flips the slope back to flat above 35.



Step 3. Together: the bull spread. Profit is capped at 3, loss is capped at 2.



Step 4. Different idea: a straddle (call + put at the same strike) is a V. It wins on big moves either way.

What it means

- **Bull call spread:** buy a low-strike call, sell a high-strike call. Cheaper than a call alone, but profit is capped. You win if the price rises.
- **Bear spread:** the opposite, for a falling price.
- **Straddle:** buy a call and a put at the same strike. You win if the price moves a lot either way. A bet on volatility.
- **Strangle:** like a straddle with two different strikes. Cheaper, needs a bigger move.
- **Butterfly:** buy calls at K1 and K3, sell two at K2. Wins if the price stays near K2 (Class "Volatility smiles" uses it to find probabilities).
- **Covered call / protective put:** stock + short call, or stock + long put (insurance).
- **Put-call parity:** a call plus cash $K \cdot e^{-rT}$ pays the same as a put plus the stock, so they must cost the same. Only for European options.
- **Bounds:** a call is worth at least $S - Ke^{-rT}$ and at most S.

The formulas

```
Put-call parity:  c + K·e^(-rT) = p + S0
Call bounds:      max(S0 - K·e^(-rT), 0) ≤ c ≤ S0
Put bounds:       max(K·e^(-rT) - S0, 0) ≤ p ≤ K·e^(-rT)
```

Profit of a strategy = sum of the payoffs − net premium paid.

Worked example

Question. (a) Buy a $30 call for $3 and sell a $35 call for $1. Max profit and max loss? (b) A European call with K = 30, T = 3 months costs $3. S = $31, r = 10%. What must the put cost?

1. (a) Net cost = 3 − 1 = 2. Max loss = 2 (price ends below 30).
2. Max payoff = 35 − 30 = 5, so max profit = 5 − 2 = 3 (price ends above 35).
3. (b) p = c + K·e^(−rT) − S = 3 + 30·e^(−0.025) − 31
4. = 3 + 29.259 − 31 = 1.26

Answer: Spread: max profit 3, max loss 2. Put = $1.26

Calculator example

Inputs: Call price c = 3, Stock S0 = 31, Strike K = 30, Rate r (%) = 10, Years T = 0.25

```
PV of strike = 30 × e^(-0.1 × 0.25) = 29.2593
Put = 3 + 29.2593 - 31 = 1.2593
Call lower bound = max(S - PV(K), 0) = 1.7407
```

Common traps

- Using put-call parity for American options (it only gives bounds there).
- Forgetting to subtract the premium when drawing profit.
- Mixing up which leg is bought and which is sold in a spread.

Where to read more

- Hull 8e, Chapters 10–11 (Properties of options, trading strategies)
- Valuation/03 Lectures/FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf

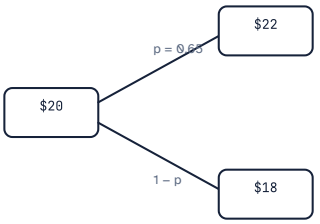
Binomial trees for options

Let the stock only go up or down each step. Then you can copy the option exactly with stock and cash, and that copy's cost is the option price.

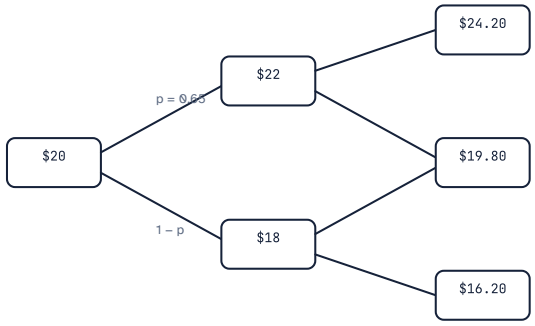
The sketch, step by step



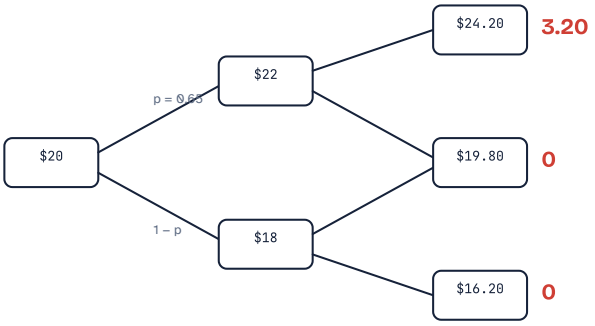
Step 1. Today the stock is $20.



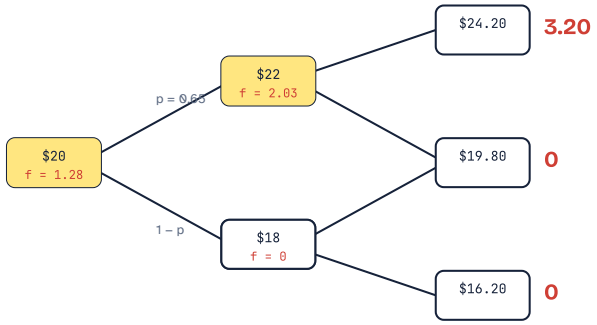
Step 2. After 3 months it is either $22 (up 10%) or $18 (down 10%).



Step 3. After 6 months: $24.20, $19.80 or $16.20.



Step 4. At the end, a call with strike $21 pays 3.20, 0, 0.



Step 5. Step back using p = 0.6523 and discount: 2.03 at the up node, then $1.28 today.

What it means

- **Replication:** hold Δ shares and borrow some cash so the portfolio pays the same as the option in both the up and down state. $\Delta = (f_u - f_d)/(S_u - S_d)$.
- **Risk-neutral probability p:** the made-up probability that makes the stock earn the risk-free rate. It is a pricing tool, not anyone's real forecast.
- **Backward induction:** start at the end with payoffs, step back one node at a time: value = $e^{-r\Delta t}[p \cdot f_u + (1-p) \cdot f_d]$.
- **American options:** at each node, value = max(hold value, exercise now). Matters most for puts.
- With many small steps the tree converges to Black-Scholes (Class 17). Typical choice: $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$.

The formulas

```
p = (e^(rΔt) - d) / (u - d)
f = e^(-rΔt) × [p·f_u + (1 - p)·f_d]
Δ = (f_u - f_d) / (S·u - S·d)
American: f = max(f, exercise value)
```

u, d = up and down factors, Δt = length of one step, f = option value at a node.

Worked example

Question. Stock $20, u = 1.1, d = 0.9, r = 12%, each step is 3 months. Price a European call with strike $21: one step, then two steps.

1. p = (e^(0.12×0.25) − 0.9) / 0.2 = (1.0305 − 0.9)/0.2 = `0.6523`
2. One step: up price 22 → payoff 1; down 18 → payoff 0. c = e^(−0.03) × 0.6523 × 1 = `0.633` (Δ = 1/4 = 0.25 shares)
3. Two steps: ends 24.2, 19.8, 16.2 → payoffs 3.2, 0, 0
4. Up node: e^(−0.03) × 0.6523 × 3.2 = `2.0257`; down node = 0
5. Today: e^(−0.03) × 0.6523 × 2.0257 = `1.2822`

Answer: One-step call = $0.633; two-step call = $1.2822

Calculator example

Inputs: Stock S = 20, Up factor u = 1.1, Down factor d = 0.9, Rate r (%) = 12, Step length (years) = 0.25, Strike K = 21

```
p = (1.0305 - 0.9) / (1.1 - 0.9) = 0.6523
Call payoffs: up 1.00, down 0.00 → call = 0.6330, Δ = 0.2500
Put payoffs: up 0.00, down 3.00 → put = 1.0124
```

Common traps

- Using the real-world probability instead of p.
- Discounting by the whole T at each step instead of Δt.
- For American puts, forgetting to compare with early exercise at every node.

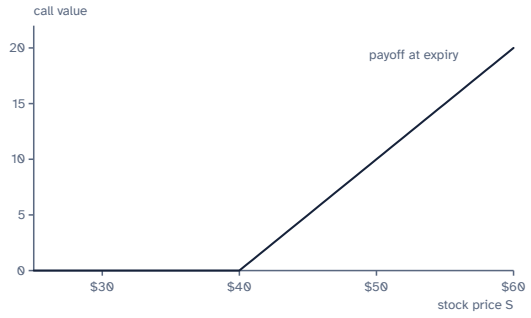
Where to read more

- `Derivatives/03 Lectures/FRE6073_LECTURE_01_Binomial_trees.pdf`
- `Hull 8e, Chapter 12`

Black-Scholes and the Greeks

Black-Scholes gives a formula for a European option price from five inputs. The Greeks say how the price moves when one input moves.

The sketch, step by step



Step 1. At expiry a call with strike $40 is worth max(S − 40, 0): a hockey stick.



Step 2. Before expiry the price is a smooth curve above the stick. The gap is time value.



Step 3. At S = $42 with 6 months left, Black-Scholes says $4.76.



Step 4. The slope of the curve there is delta = 0.78. Move the stock $1, the call moves about $0.78.



Step 5. With 3 months left (dashed) the curve sits lower and hugs the stick. That loss over time is theta. How sharply the curve bends is gamma.

What it means

Idea behind it: hold the option and short Δ shares. Over a tiny instant the randomness cancels, so the hedge must earn the risk-free rate. That gives a PDE whose answer is the formula. Prof. Shimko's "Simpler Option Pricing" lecture reaches the same formula with only college calculus.

- **Inputs:** stock S, strike K, rate r, volatility σ, time T. Note: the expected return μ is *not* an input.
- **Delta** N(d1): change in option price per $1 move in the stock. Also the hedge ratio.
- **Gamma:** how fast delta changes. High near the strike close to expiry.

- **Theta:** value lost per day as time passes. **Vega:** change per 1% more volatility. **Rho:** change per 1% higher rate.
- **Put-call parity:** $c + Ke^{-rT} = p + S$. Use it to check your answer.

The formulas

```
c = S·N(d1) - K·e^(-rT)·N(d2)
p = K·e^(-rT)·N(-d2) - S·N(-d1)
d1 = [ln(S/K) + (r + σ²/2)·T] / (σ√T)
d2 = d1 - σ√T
Delta(call) = N(d1)    Gamma = φ(d1)/(S·σ·√T)    Vega = S·φ(d1)·√T
```

N = cumulative normal (NORM.S.DIST(x,TRUE) in Excel), φ = normal density.

Worked example

Question. S = 42, K = 40, r = 10%, σ = 20%, T = 0.5 years (Hull's example). Find the call, the put and delta.

1. σ√T = 0.2 × 0.7071 = `0.1414`
2. d1 = [ln(1.05) + (0.10 + 0.02) × 0.5] / 0.1414 = (0.0488 + 0.06)/0.1414 = `0.7693`
3. d2 = 0.7693 − 0.1414 = `0.6278`; N(d1) = 0.7791, N(d2) = 0.7349
4. c = 42 × 0.7791 − 40 × e^(−0.05) × 0.7349 = `4.76`
5. Put by parity: p = 4.76 − 42 + 40e^(−0.05) = `0.81`. Delta = `0.779`

Answer: Call $4.76, put $0.81, delta 0.78

Calculator example

Inputs: Stock S = 42, Strike K = 40, Rate r (%) = 10, Volatility σ (%) = 20, Years T = 0.5

```
d1 = 0.7693, d2 = 0.6278
N(d1) = 0.7791, N(d2) = 0.7349
Call = 4.7594, Put = 0.8086
Delta = 0.7791, Gamma = 0.0500, Vega (per 1%) = 0.0881
```

Common traps

- Using σ^2 where σ goes, or forgetting √T.
- Using a non-continuous rate in e^(−rT).
- Thinking the stock's expected return changes the price. It does not.

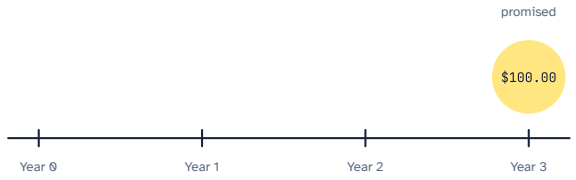
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_12_Simpler_Option_Pricing_Update_2026.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf`
- `Hull 8e, Chapters 14 and 18`

Money today vs money later

A dollar later is worth less than a dollar today, because today's dollar could sit in a bank and grow. Present value (PV) just undoes that growth.

The sketch, step by step



Step 1. This is a timeline. Today is year 0. Someone promises you $100 at year 3.



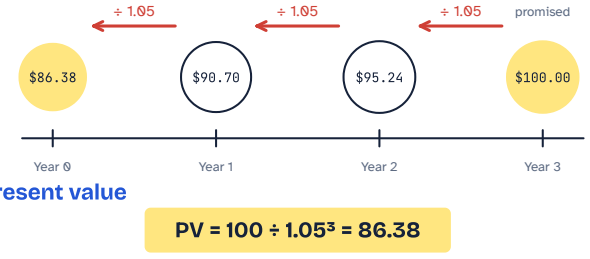
Step 2. Move the $100 back one year: divide by 1.05. It shrinks to $95.24.



Step 3. Back one more year: divide by 1.05 again. Now $90.70.



Step 4. Back to today: $86.38. That is the present value. Smaller coin, same "worth".



Step 5. Shortcut: do all three divides at once. PV = 100 ÷ 1.05^3 = 86.38.

What it means

Think of the bank as a time machine for money. Put $\$V_0$ in at 5% and next year you get $\$V_0 \times 1.05$. Present value runs the machine backwards: divide instead of multiply.

- **Future value (FV):** what money grows into. Multiply by (1 + r) each year.
- **Present value (PV):** what future money is worth today. Divide by (1 + r) each year.
- **Discount factor:** the number you multiply a future cash flow by to get its PV. For 3 years at 5% it's $1/1.05^3 = 0.8638$.
- **Annuity:** the same payment every period. **Perpetuity:** the same payment forever, PV = C / r.

- Prof. Shimko's point ("PV fails"): PV only works cleanly when you know the cash flows and the right rate. Risky cash flows need a risk adjustment, which later classes cover.

The formulas

```
PV = CF / (1 + r)^t
Perpetuity:  PV = C / r
Annuity:     PV = C × [1 - (1+r)^(-n)] / r
```

CF = cash flow, r = rate per period, t = number of periods, C = repeating payment.

Worked example

Question. You will get $100 in 3 years. The bank pays 5% a year. What is it worth today?

1. Write the formula: `PV = 100 / 1.05^3`
2. Work out $1.05^3 = 1.05 \times 1.05 \times 1.05 =$ `1.157625`
3. Divide: `100 / 1.157625 = 86.38`
4. Check by going forward: 86.38 × 1.157625 = 100. It works.

Answer: PV = $86.38

Calculator example

Inputs: Future cash flow ($) = 100, Rate per period (%) = 5, Periods = 3

```
Discount factor = 1 / 1.0500^3 = 0.8638
PV = 100.00 × 0.8638 = $86.38
```

Common traps

- Using an annual rate with monthly cash flows. Match the rate to the period (6% a year = 0.5% a month).
- Mixing up multiply (future) and divide (present).
- Forgetting that a perpetuity's first payment comes one period from now, not today.

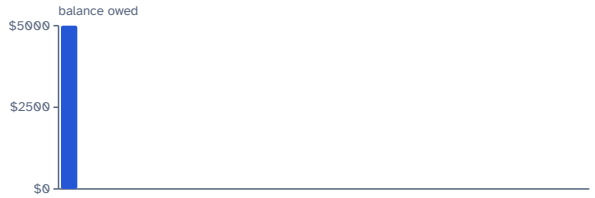
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_01_Lecture_1_Present_Value_Fails.pdf`
- `Valuation/02 Books and Readings/FRE6103_BOOK_03_Valuation_Principles_Ch_1_2.pdf`

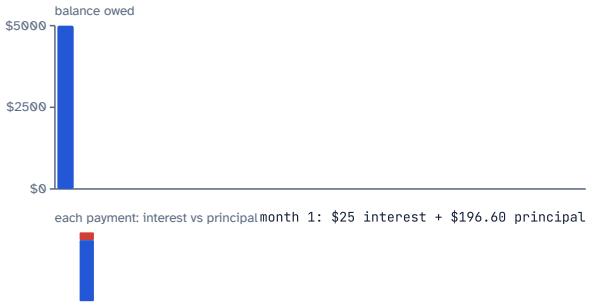
Loans: payments, default and prepayment

A loan payment is split into interest first, then the rest pays down what you owe. A pool of loans also loses money to people who default or pay off early.

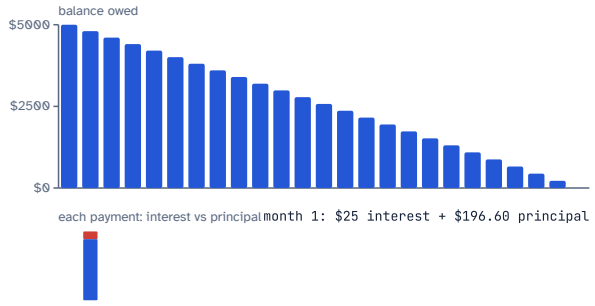
The sketch, step by step



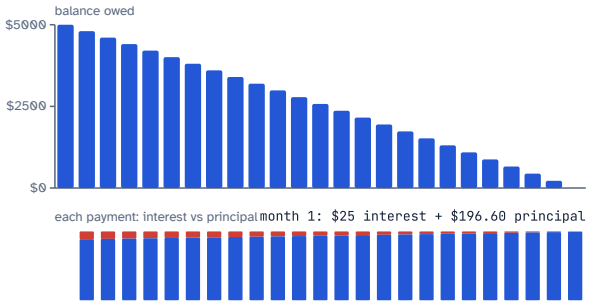
Step 1. You borrow $5,000. That tall bar is what you owe at month 0.



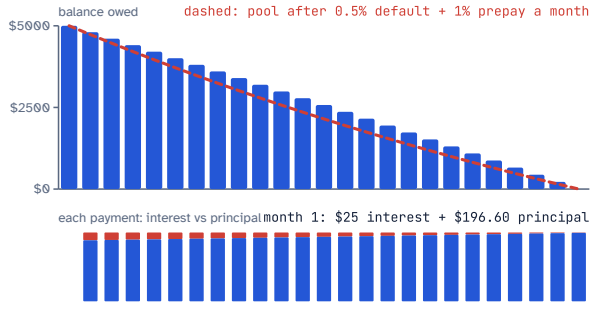
Step 2. Each month you pay $221.60. Red part = interest, blue part = paying down the loan.



Step 3. Over 24 months the balance falls to zero.



Step 4. Notice the red part shrinks and the blue part grows: less balance means less interest.



Step 5. For a lender with a pool of loans, defaults and prepayments make the pool (dashed) shrink faster.

What it means

This is the lecture's own example: borrow $5,000 for 24 months at 6% a year. Every month you pay the same amount. Early on most of it is interest, later most of it is principal, because the balance (and so the interest) keeps shrinking.

- **Amortization:** paying the loan down to zero with equal payments.
- **Borrower's view:** what does it cost me? **Lender's view:** what cash do I actually get back?
- **Default:** some borrowers stop paying. In the course model, a fixed share of the remaining pool defaults each month (0.5% in Mini Project 1).

- **Prepayment:** some borrowers pay everything early (1% a month in the project). The lender gets cash sooner but loses future interest.

The formulas

```
Payment:   PMT = B × i / [1 - (1+i)^(-n)]
Each month: interest = balance × i
            principal paid = PMT - interest
            new balance = balance - principal paid
Pool left after k months ≈ (1 - d - p)^k
```

B = amount borrowed, i = monthly rate (6%/12 = 0.5%), n = number of months, d = monthly default share, p = monthly prepay share.

Worked example

Question. Borrow $5,000 for 24 months at 6% a year. What is the payment, and what happens in month 1?

1. Monthly rate i = 6% / 12 = `0.005`
2. PMT = 5000 × 0.005 / (1 – 1.005^(–24)) = `$221.60` (Excel: PMT(0.06/12,24,–5000))
3. Month 1 interest = 5000 × 0.005 = `$25.00`
4. Principal paid = 221.60 – 25.00 = `$196.60`, new balance = `$4,803.40`

Answer: Payment $221.60; balance after month 1 is $4,803.40

Calculator example

Inputs: Amount borrowed ($) = 5000, Annual rate (%) = 6, Months = 24

```
Monthly rate = 0.5000%
Payment = $221.60
Month 1: interest $25.00, principal $196.60, balance $4,803.40
Total paid = $5,318.47, so total interest = $318.47
```

Common traps

- Using 6% instead of 0.5% for the month.
- Forgetting interest is paid before principal.
- In the pool model, default and prepayment both shrink the balance, but only default loses money.

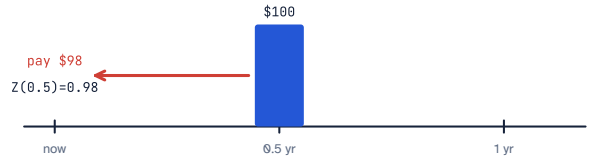
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_03_Lecture_2A_Loans_with_exponential_default_risk.pdf`
- `Valuation/07 Earlier Midterm Materials/FRE6103_REVIEW_02_Mini_Project_1_Two_Sides_of_a_Loan_Instructions.txt`

Bonds and building a zero curve

A zero-coupon bond pays one amount at one date, so its price tells you today's value of $1 at that date. Bootstrapping peels these discount factors out of coupon bonds one date at a time.

The sketch, step by step



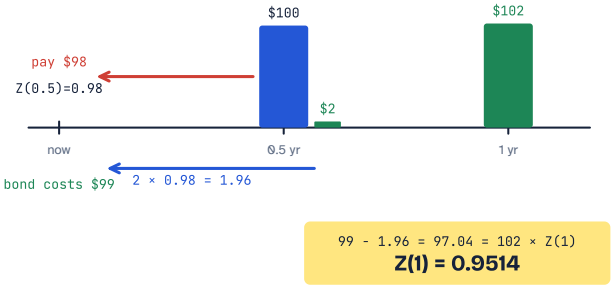
Step 1. Start with the easy one. A 6-month zero costs $98 for $100. So $1 in 6 months is worth $0.98 today.



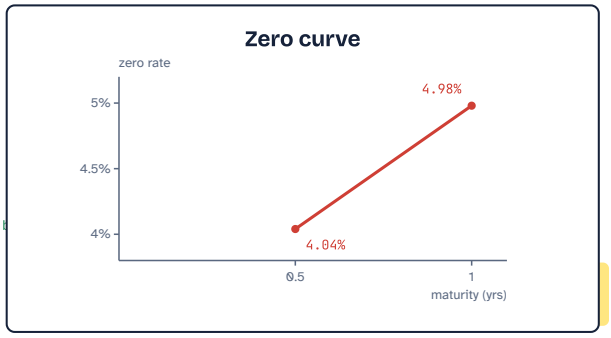
Step 2. Next, a 1-year coupon bond costing $99. It pays $2 at 6 months and $102 at 1 year.



Step 3. We already know how to value the $2: 2 × 0.98 = $1.96.



Step 4. The rest of the price must be the $102: 99 – 1.96 = 97.04. So Z(1) = 97.04 / 102 = 0.9514.



Step 5. Turn each Z into a rate and plot. That is your zero curve.

What it means

A coupon bond is just a bundle of little zero-coupon bonds, one for each payment. If you already know the discount factor for the early dates, you can subtract those pieces from the bond price. What's left must be the value of the last payment, which gives you the next discount factor. Repeat. That is **bootstrapping**.

- **Discount factor Z(t):** price today of $1 paid at time t.
- **Zero rate:** the rate that matches Z(t). Continuous: r = –ln Z(t) / t.
- **Zero curve / term structure:** zero rates plotted against maturity. Mini Project 2 builds this from Treasury data.

The formulas

```
Bond price = Σ CF(t) × Z(t)
Z(t) = e^(-r·t)      →   r = -ln Z(t) / t
Bootstrap last date:  Z(T) = (Price - Σ earlier CF × Z) / CF(T)
```

CF(t) = cash flow at time t, Z(t) = discount factor, r = zero rate (continuous).

Worked example

Question. A 6-month zero sells at $98 per $100. A 1-year bond with a 4% coupon paid twice a year ($2 every 6 months) sells at $99. Find Z(0.5) and Z(1).

1. Zero bond: `Z(0.5) = 98 / 100 = 0.98`
2. The 1-year bond pays $2 at 0.5 years and $102 at 1 year.
3. Value of the first coupon: `2 × 0.98 = 1.96`
4. What is left: `99 - 1.96 = 97.04 = 102 × Z(1)`, so `Z(1) = 0.9514`
5. Zero rates: −ln(0.98)/0.5 = `4.04%`, −ln(0.9514)/1 = `4.98%`

Answer: Z(0.5) = 0.98, Z(1) = 0.9514; zero rates 4.04% and 4.98%

Common traps

- Forgetting the final payment includes the coupon (102, not 100).
- Using the annual coupon (4) instead of the half-year coupon (2).
- Mixing continuous and annual zero rates.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_08_Treasury_data_090426.xlsx`
- `Derivatives/03 Lectures/FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx`

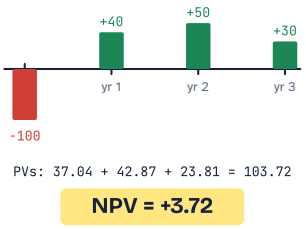
Capital budgeting: NPV and IRR

A company should take a project if its NPV is positive: the present value of what it brings in is more than what it costs. IRR is the discount rate where NPV is exactly zero.

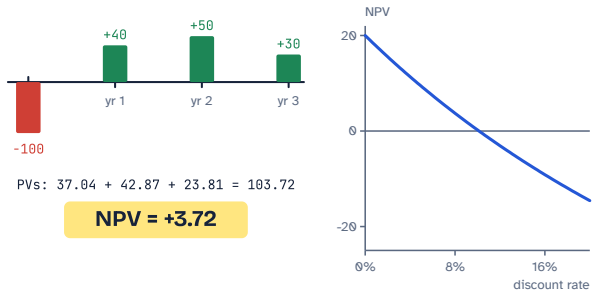
The sketch, step by step



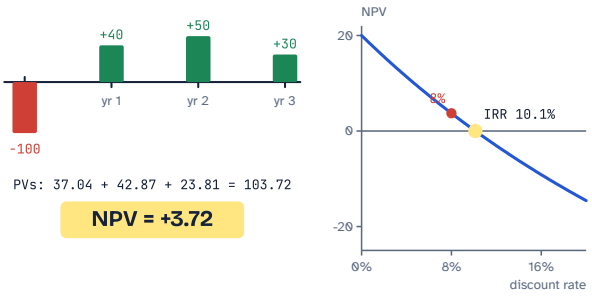
Step 1. Cash flows on a timeline: pay $100 now, get 40, 50, 30 later.



Step 2. Shrink each future amount to today's money at 8%, then add. NPV = +3.72.



Step 3. Now try every discount rate. NPV falls as the rate rises.



Step 4. Where the curve crosses zero (about 10.1%) is the IRR. Our 8% is to the left, so NPV > 0.

What it means

- **NPV:** add up the PV of every cash flow, including the (negative) cost today.
- **Discount rate:** the cost of capital, i.e. the return investors could get elsewhere at the same risk (CAPM from the Portfolio class helps here).
- **IRR:** the rate that makes NPV = 0. If IRR > cost of capital, NPV is positive. Find it by trial and error or Excel's IRR().
- **When IRR misleads:** projects of different size, cash flows that change sign more than once (can have two IRRs), or different timing. When in doubt, trust NPV.
- **Payback period** ignores timing and anything after the cutoff. Use it only as a rough check.
- Welch's book also covers **capital structure**: how mixing debt and equity changes the cost of capital (WACC).

The formulas

```
NPV = Σ CF_t / (1 + r)^t        (t = 0, 1, 2, …)
IRR: the r where NPV = 0
WACC = E/V × rE + D/V × rD × (1 - tax)
```

CF_0 is usually negative (the investment). E, D = market values of equity and debt, V = E + D.

Worked example

Question. A project costs $100 today and pays $40, $50 and $30 at the end of years 1, 2 and 3. Cost of capital 8%. NPV? IRR?

1. PV of year 1: 40 / 1.08 = `37.04`
2. PV of year 2: 50 / 1.08^2 = `42.87`
3. PV of year 3: 30 / 1.08^3 = `23.81`
4. NPV = 37.04 + 42.87 + 23.81 − 100 = `3.72` → take it
5. IRR (Excel IRR or trial): about `10.1%`, above 8%. Same answer.

Answer: NPV = $3.72, IRR ≈ 10.1%

Calculator example

Inputs: Cost today = 100, Year 1 = 40, Year 2 = 50, Year 3 = 30, Discount rate (%) = 8

```
NPV at 8% = 3.72 → take it
IRR ≈ 10.13%
```

Common traps

- Forgetting the year-0 cost.
- Choosing between two projects by IRR instead of NPV.
- Using a company-wide discount rate for a project with very different risk.

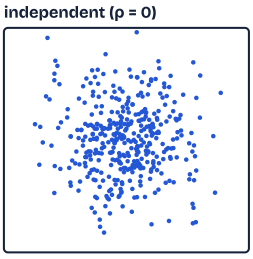
Where to read more

- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_13_VL10_Multiperiod_cash_flows.pdf`

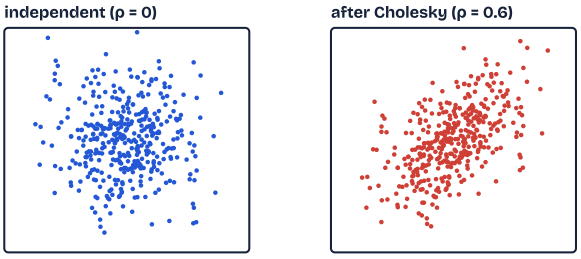
Simulation, correlation and Cholesky

Simulation means making up thousands of possible futures with random numbers and averaging the results. Cholesky is the trick that makes those random numbers move together.

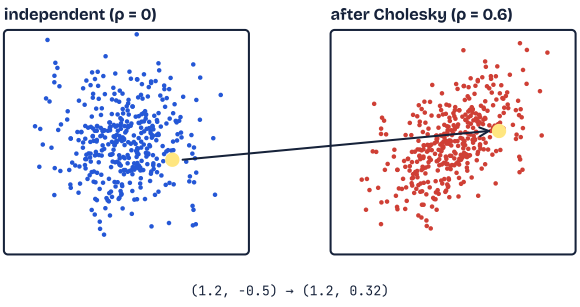
The sketch, step by step



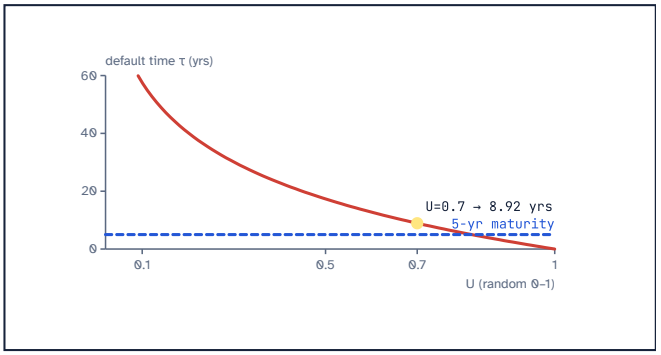
Step 1. 400 pairs of independent random numbers. A round cloud: knowing one tells you nothing about the other.



Step 2. Apply Cholesky with ρ = 0.6. The cloud stretches into a tilted oval. Now they move together.



Step 3. One dot: (Z1, Z2) = (1.2, −0.5) becomes (1.2, 0.32). It moved up because Z1 was high.



Step 4. Default times: draw U, then τ = −ln(U)/λ. Small U means a long wait; big U means early default.

What it means

Some problems are too messy for a formula (10 bonds defaulting at different times, a CDO waterfall). So you let the computer roll dice many times and look at the outcomes.

- **Uniform U** (between 0 and 1) → **normal Z** with NORM.S.INV(U) in Excel.
- **Correlation:** independent Z's don't move together. To get correlation ρ, mix them: X2 = ρ·Z1 + √(1−ρ²)·Z2. For many variables, the Cholesky matrix does this mixing for all of them at once.
- **Default time:** with a constant default rate λ, a random default time is τ = −ln(U)/λ. If τ is before maturity, that bond defaults in this run.
- **Sanity checks:** the sample mean, standard deviation and correlation should match what you asked for. "Moment matching" forces them to match exactly.

The formulas

```
Two variables:   X1 = Z1
                 X2 = ρ·Z1 + √(1 - ρ²)·Z2
Many variables:  X = L·Z   where  L·Lᵀ = correlation matrix
Default time:    τ = -ln(U) / λ
```

Z = independent standard normals, L = Cholesky (lower triangular) matrix, λ = default intensity per year.

Worked example

Question. ρ = 0.6, and you drew Z1 = 1.2, Z2 = −0.5. Find X1 and X2. Then: λ = 4% a year and U = 0.7. When does the bond default?

1. X1 = Z1 = `1.2`
2. √(1 − 0.36) = `0.8`
3. X2 = 0.6 × 1.2 + 0.8 × (−0.5) = 0.72 − 0.40 = `0.32`
4. τ = −ln(0.7)/0.04 = 0.3567/0.04 = `8.92 years`. For a 5-year bond: no default in this run.

Answer: X1 = 1.2, X2 = 0.32; default at 8.92 years

Common traps

- Using a covariance matrix when the formula needs a correlation matrix (or the other way round).
- Re-drawing random numbers every time Excel recalculates. Fix them (paste values) so results are comparable.
- Forgetting to check that sample correlation is near the target.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_05_Lecture_4_Simulation.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_02_Lecture_4_Cholesky_five.xlsx`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_03_Lecture_4_Simulation_spreadsheets.xlsx`

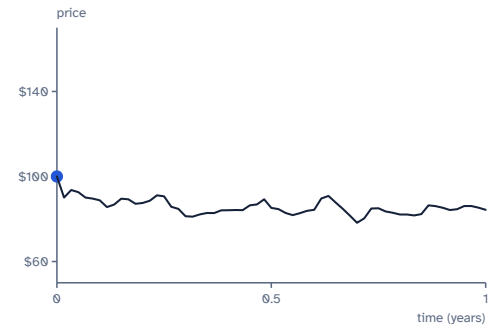
Random walks and stock price paths

A stock price is modelled as a steady drift plus random shocks. Using percentage shocks (geometric Brownian motion) keeps prices positive and gives a lognormal shape.

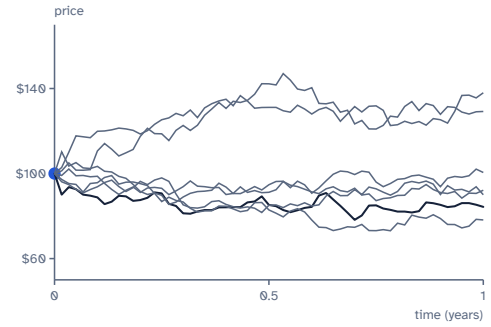
The sketch, step by step



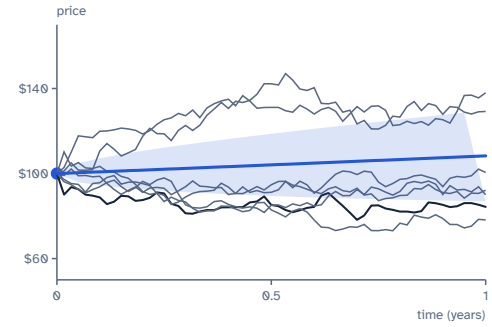
Step 1. Start at $100. Time runs left to right over one year.



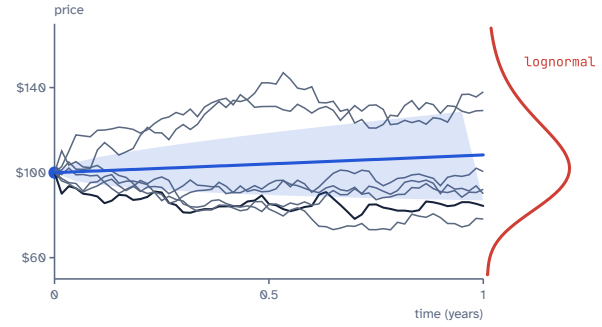
Step 2. One simulated path: small random up and down steps with a slight upward drift.



Step 3. More paths. Each one is a possible future.



Step 4. The blue line is the average path, 100·e^(0.08t). The band shows the typical spread, which widens over time.



Step 5. At year end the prices pile up in a lopsided (lognormal) shape: a long tail upward, never below zero.

What it means

- **Wiener process (Brownian motion) W:** a random walk where each tiny step is normal with variance equal to the time step. Over time Δt the change is ε√Δt.
- **Arithmetic Brownian motion:** $dX = \mu\,dt + \sigma\,dW$. Can go negative. Fine for spreads or rates, not for stock prices.
- **Geometric Brownian motion (GBM):** $dS = \mu S\,dt + \sigma S\,dW$. Returns are normal, prices are lognormal.
- **Simulating one step exactly:** $S_{next} = S \times \exp((\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}\cdot Z)$. The $-\sigma^2/2$ is there because of Itô's lemma; without it your average would be too high.
- **Mean reversion** (e.g. rates, commodities): the drift pulls the value back toward a long-run level.

The formulas

```
GBM:          dS = μ·S·dt + σ·S·dW
One step:     S(t+Δt) = S(t) × exp[(μ - σ²/2)·Δt + σ·√Δt·Z]
Expected:     E[S_T] = S0 × e^(μT)
```

μ = expected return (drift), σ = volatility, Z = standard normal draw.

Worked example

Question. S = 100, μ = 8%, σ = 20%, Δt = 1 year, and you draw Z = 0.5. What is the simulated price?

1. Drift part: (0.08 − 0.02) × 1 = `0.06`
2. Shock part: 0.20 × √1 × 0.5 = `0.10`
3. Add: 0.16, then `e^0.16 = 1.1735`
4. S = 100 × 1.1735 = `117.35`

Answer: S = $117.35

Common traps

- Forgetting $-\sigma^2/2$ in the exponent.
- Using σ × Δt instead of σ × √Δt.
- Mixing up the median (S0·e^((μ−σ²/2)T)) and the mean (S0·e^(μT)).

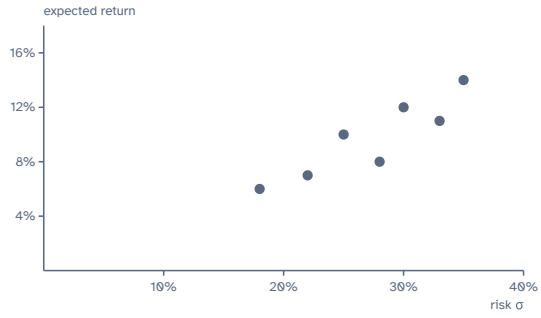
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_06_Lecture_5_Stochastic_Processes.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_05_Lecture_5_Simulation_spreadsheets.xlsx`
- `Hull 8e, Chapter 13 (Wiener processes and Itô)`

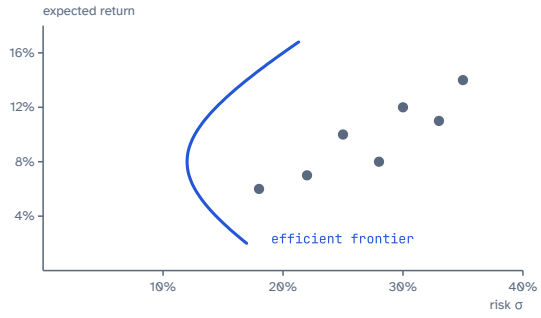
Portfolio theory and the CAPM

Investors only get paid for risk they cannot diversify away. Beta measures that risk, and the CAPM turns beta into a required return.

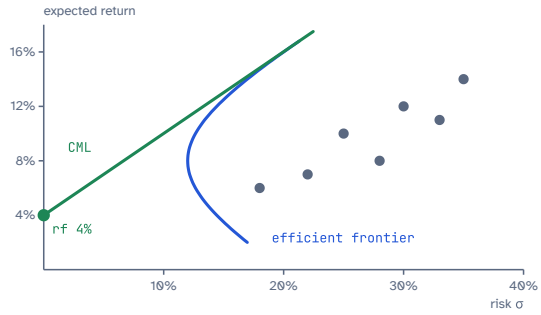
The sketch, step by step



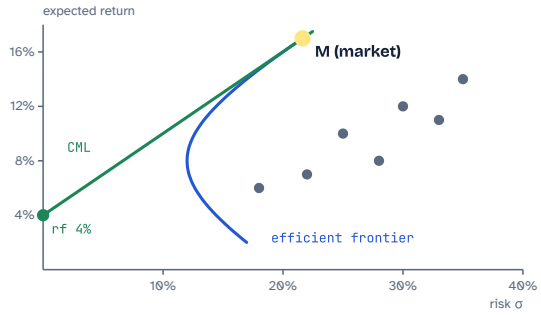
Step 1. Each dot is one stock: its risk (σ) and expected return.



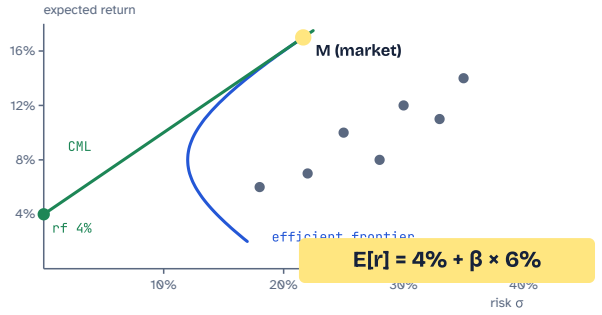
Step 2. Mixing them gives the efficient frontier: the best return for each level of risk.



Step 3. Add the risk-free asset at 4%. Draw a line from it that just touches the frontier.



Step 4. The touching point is the market portfolio M. Everyone should hold a mix of M and the risk-free asset.



Step 5. For one asset, CAPM says: required return = 4% + β × 6%.

What it means

- **Efficient frontier:** the best possible return for each level of risk, built by mixing stocks (Markowitz). Correlation below 1 bends it to the left.
- **Capital market line:** mix the risk-free asset with the best risky portfolio (the "market" M). It is the line from r_f that just touches the frontier. Its slope is the Sharpe ratio.
- **Beta:** how much an asset moves when the market moves. $\beta = \text{Cov}(r_i, r_M)/\text{Var}(r_M)$.
- **CAPM:** $E[r] = r_f + \beta \times (E[r_M] - r_f)$.

- **Lintner (certainty-equivalent) form:** value a risky cash flow X by subtracting a risk charge, then discounting at the risk-free rate. This is how the course values CDO tranches.

The formulas

```
Sharpe ratio:  (E[r] - rf) / σ
CAPM:          E[r_i] = rf + β_i × (E[r_M] - rf)
Lintner:       PV = (E[X] - λ·Cov(X, r_M)) / (1 + rf)
               λ = (E[r_M] - rf) / σ_M²
```

λ = market price of risk, X = an uncertain cash flow in one period.

Worked example

Question. rf = 4%, market return 10%, σ_M = 20%. (a) Required return for β = 1.2? (b) A cash flow has E[X] = $110 and Cov(X, r_M) = 2. Value it with Lintner.

1. (a) 4% + 1.2 × (10% − 4%) = `11.2%`
2. (b) λ = 0.06 / 0.04 = `1.5`
3. Risk charge = 1.5 × 2 = `3`, certainty equivalent = 110 − 3 = 107
4. PV = 107 / 1.04 = `102.88`

Answer: Required return 11.2%; PV = $102.88

Common traps

- Using total risk σ instead of beta for required return.
- Discounting the certainty equivalent at a risky rate. It already includes the risk charge, so use rf.
- Forgetting that correlation, not volatility alone, drives diversification.

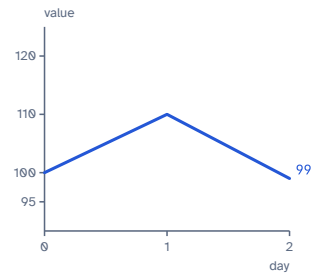
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_07_Lecture_6_MPT_and_the_CAPM.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_06_MPT_Ten_Stock_Portfolio.xlsx`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_07_sharpe_and_lintner.xlsx`

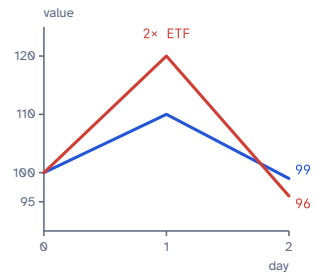
ETFs and futures for end users

Futures and ETFs look simple, but an end user faces hidden costs: rolling futures in contango, daily-reset leverage in leveraged ETFs, margin calls and basis risk.

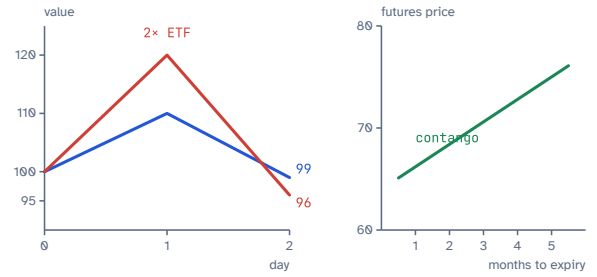
The sketch, step by step



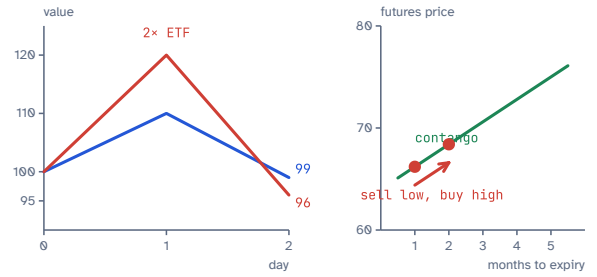
Step 1. An index that goes +10% then −10%: it ends at 99.



Step 2. A 2× daily ETF: +20% then −20%. It ends at 96, worse than "2 × −1%".



Step 3. Contango: futures prices rise with maturity. Each month the fund sells cheap near futures and buys pricier ones.



Step 4. That gap is lost every roll: a slow leak called negative roll yield.

What it means

- **ETF:** a fund that trades like a stock and tracks an index. Low cost, but commodity ETFs usually hold futures, not the commodity.
- **Roll:** futures expire, so a fund sells the near contract and buys the next one every month.
- **Contango** (futures above spot, curve upward): each roll buys higher and the fund slowly loses (negative roll yield). **Backwardation** (curve downward) gives positive roll yield.
- **Leveraged ETFs** (2×, 3×) reset every day. In a choppy market they lose value even if the index ends flat (volatility decay).
- **End-user risks:** margin/liquidity calls, basis risk, volume (volumetric) risk when you do not know how much you will need to hedge.

The formulas

```
Leveraged ETF over 2 days:  (1 + L·r1)(1 + L·r2)   ≠   1 + L·(total index return)
Roll yield (approx):         (F_near - F_next) / F_near  per roll
```

L = leverage (2 for a 2× ETF), r = daily index return.

Worked example

Question. An index goes +10% then −10%. What happens to the index and to a 2× daily ETF?

1. Index: 1.10 × 0.90 = `0.99` → down 1%
2. 2× ETF: day 1 = +20%, day 2 = −20%
3. 1.20 × 0.80 = `0.96` → down 4%
4. The ETF lost 4%, not 2× the 1% (2%). That extra loss is volatility decay.

Answer: Index −1%, 2× ETF −4%

Common traps

- Expecting a 2× ETF to give 2× the return over months.
- Ignoring roll cost when comparing a commodity ETF to the spot price.
- Forgetting that futures hedges can create big cash margin calls even when the hedge works.

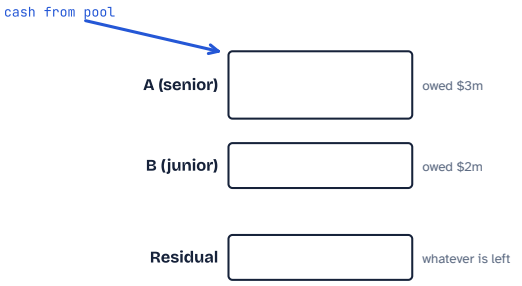
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_08_Lecture_7_ETFs_and_Futures_MARKUP.pdf`
- `Futures lecture recording (see "Watch online")`

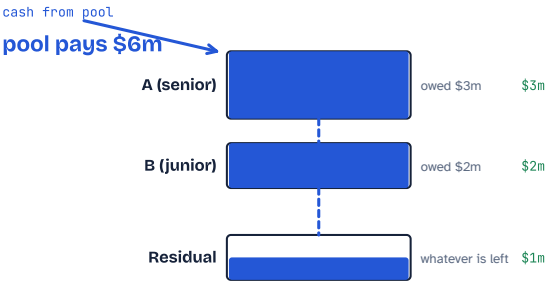
CDO waterfalls and tranches

A CDO pools many bonds and pays the cash out in order: senior tranche first, then junior, then whatever is left goes to equity. Who gets hurt depends on how many bonds default.

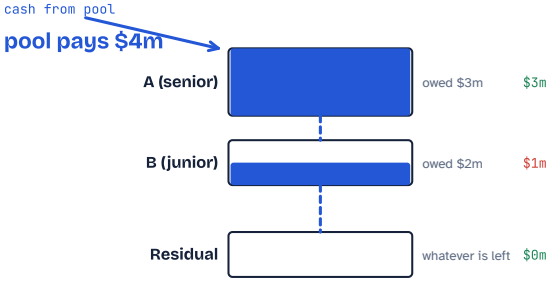
The sketch, step by step



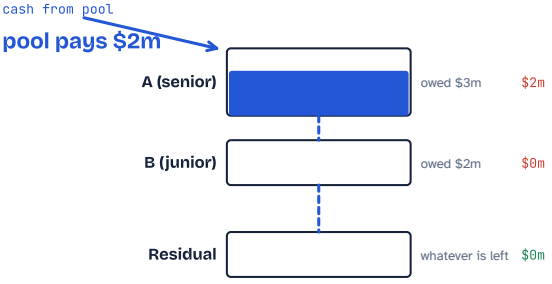
Step 1. Three buckets stacked like a fountain: A on top, B in the middle, residual at the bottom. Cash pours in from the pool.



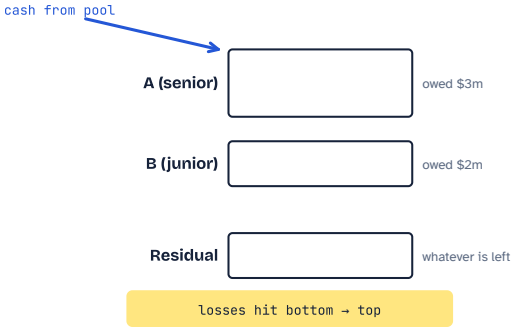
Step 2. Good period: the pool pays $6m. A fills ($3m), B fills ($2m), $1m spills into the residual.



Step 3. Some defaults: only $4m. A fills, B gets $1m of its $2m, residual gets nothing.



Step 4. Many defaults: $2m. Even A comes up short. B and residual get nothing.



Step 5. Rule: losses hit from the bottom up. Senior is protected by everyone below it.

What it means

- **Collateral:** the pool of bonds (Mini Project 3: ten $10m bonds, 6% coupons).
- **Waterfall:** each period, pay tranche A what it is owed, then B, then the residual. No carry-forwards in the project version.
- **Senior (A)** is safest; it only loses if many bonds default. **Equity/residual** takes the first losses.
- **Correlation matters:** with high correlation, bonds tend to default together. That makes "almost nothing" and "lots" of defaults both likelier. Senior tranches get *riskier*; equity can get *safer*.
- Value each tranche by simulating cash flows (Class 11) and pricing them with Lintner (Class 13).

The formulas

```
Pay A = min(cash, owed to A)
Pay B = min(cash - Pay A, owed to B)
Residual = cash - Pay A - Pay B
```

Repeat every period in every simulation run, then average and discount.

Worked example

Question. This period A is owed $3m and B is owed $2m. The pool pays (a) $6m, (b) $4m, (c) $2m. Who gets what?

1. (a) $6m: A gets 3, B gets 2, residual gets `1`
2. (b) $4m: A gets 3, B gets `1` (short by 1), residual `0`
3. (c) $2m: A gets `2` (short by 1), B `0`, residual `0`
4. So the residual is hit first, then B, and A only in bad cases.

Answer: (a) 3/2/1 (b) 3/1/0 (c) 2/0/0

Common traps

- Paying tranches pro rata instead of in order.
- Forgetting LGD: a defaulted bond still pays its recovery.
- Assuming correlation always makes every tranche riskier.

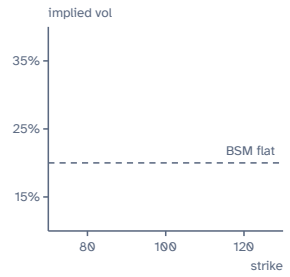
Where to read more

- `Valuation/07 Earlier Midterm Materials/FRE6103_REVIEW_04_Mini_Project_3_CDO_Risk_Part_1_Instructions.txt`
- `Valuation/06 Project and Recording Notes/FRE6103_NOTES_01_Project_and_recording_notes.md`
- `Valuation/03 Lectures/FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf`

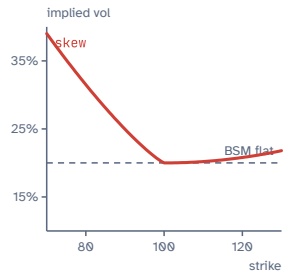
Volatility smiles and implied densities

If you back out σ from real option prices, it is not the same for every strike. That pattern tells you what the market really thinks future prices look like.

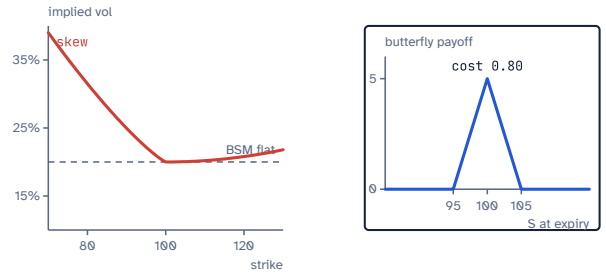
The sketch, step by step



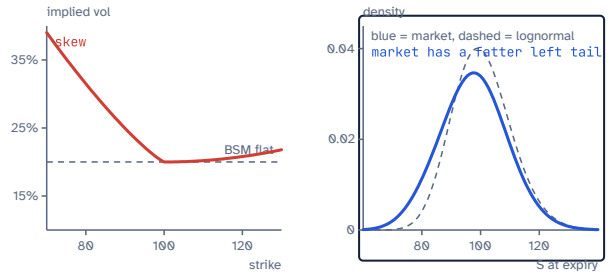
Step 1. Black-Scholes assumes one volatility for all strikes: a flat line.



Step 2. Real markets show a skew: low strikes (crash protection) have much higher implied vol.



Step 3. A butterfly (buy 95, sell two 100s, buy 105) pays a little tent. Its cost measures probability near 100.



Step 4. Line up many butterflies and you get the implied density. Compared with the lognormal (dashed), the left tail is fatter.

What it means

- **Implied volatility:** the σ you plug into Black-Scholes to match a market price. Solve it with Goal Seek.
- **Smile / skew:** plot implied vol against strike. For stocks, low strikes usually have higher vol (a skew): crashes are feared more than the lognormal model allows.
- **Volatility surface:** implied vol across strike *and* maturity.
- **Implied density (Breeden-Litzenberger):** the second derivative of call price with respect to strike gives the market's risk-neutral probability density. In practice: a butterfly spread.
- **European vs Asian:** an Asian option pays on the average price, which is less volatile, so it is cheaper. It usually needs simulation.

The formulas

$$\text{Implied density:}\quad q(K) = e^{rT} \times \partial^2 C/\partial K^2$$

$$\text{Butterfly approx:}\quad q(K) \approx e^{rT} \times [C(K-h) - 2\cdot C(K) + C(K+h)] / h^2$$

C(K) = call price at strike K, h = gap between strikes.

Worked example

Question. Calls cost 8.00, 5.00 and 2.80 at strikes 95, 100, 105. Estimate the probability density at 100 (ignore discounting). Roughly what is the chance S_T ends between 97.5 and 102.5?

1. Butterfly cost: 8.00 − 2 × 5.00 + 2.80 = `0.80`
2. Divide by $h^2 = 25$: density ≈ `0.032` per $1
3. Over a $5 window: 0.032 × 5 = `0.16`
4. So about a 16% risk-neutral chance of ending near $100.

Answer: Density ≈ 0.032 per $; about 16% chance

Common traps

- Reading a risk-neutral density as a real-world forecast.
- A negative butterfly cost means arbitrage or bad data, not a negative probability.
- Using one σ for every strike when the smile is steep.

Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_15_Volatility_Surface.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf`
- `Valuation/04 Spreadsheets and Data/FRE6103_DATA_01_European_and_Asian_option_benchmarking.xlsx`

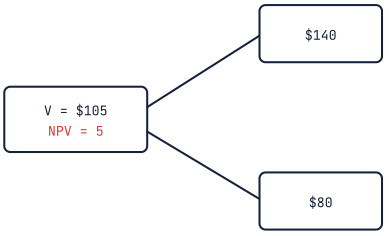
Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

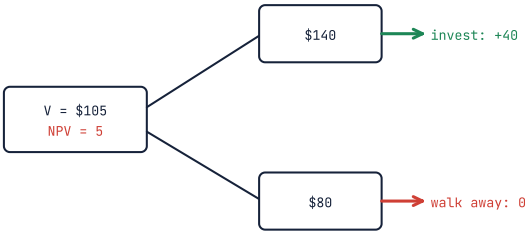
The sketch, step by step



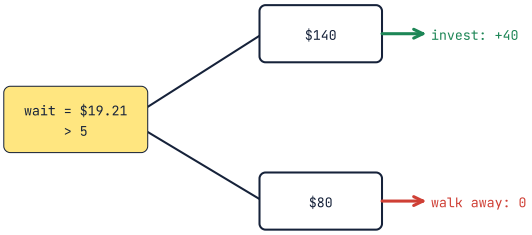
Step 1. Today the project is worth $105 and costs $100. NPV = $5.



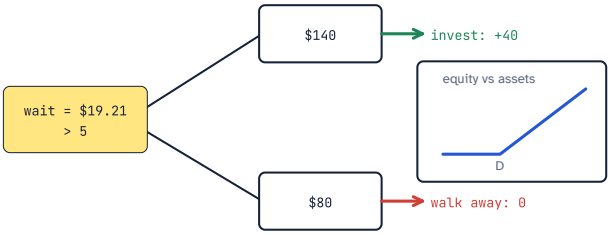
Step 2. Next year it is worth $140 (good) or $80 (bad).



Step 3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.



Step 4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.



Step 5. Same idea for a firm: shareholders get max(Assets – Debt, 0). Equity is a call on the assets.

What it means

- **Real option:** the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.
- Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.
- If the option value is bigger than investing now (NPV today), **wait**.
- **Energy example (VL14):** a power plant is a call on the spread between power and fuel prices: run it only when profitable.
- **Merton model:** equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. **Distance to default** counts how many σ the firm is from that line.

The formulas

```
Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)
Option value   = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)
Merton:  Equity = V·N(d1) - D·e^(-rT)·N(d2)
         DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T),   PD = N(-DD)
```

V = value of project or firm assets, I = investment cost, D = face value of debt.

Worked example

Question. A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 − 100 = `5`
2. p = (105 × 1.05 − 80) / (140 − 80) = 30.25/60 = `0.504`
3. Wait: next year invest only if up: payoff 40 or 0
4. Option value = 0.504 × 40 / 1.05 = `19.21`. Bigger than 5, so wait.
5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = `1.01`, PD = N(−1.01) ≈ `15.7%`

Answer: Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common traps

- Using the real-world probability in the roll-back.
- Comparing the option with zero instead of with investing now.
- In Merton, using the stock's volatility as the asset volatility.

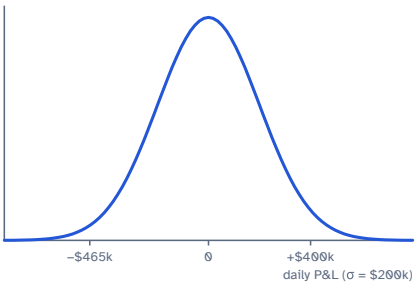
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf`
- `Valuation/03 Lectures/FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf`
- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`
- `Risk syllabus RMN6 (Merton, distance to default)`

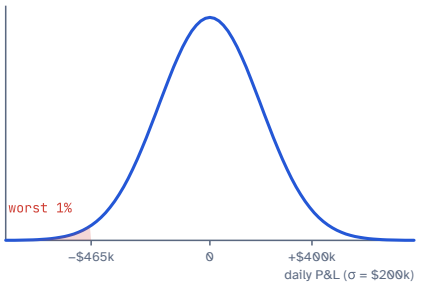
Volatility, diversification, VaR and ES

VaR answers "how bad is a bad day?" Expected Shortfall answers "and if it gets worse than that, how bad on average?"

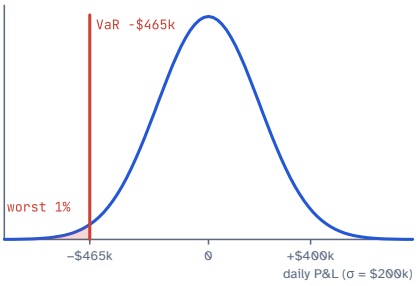
The sketch, step by step



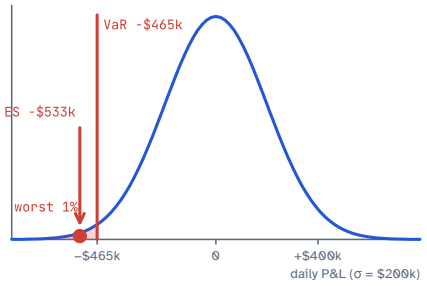
Step 1. This bell curve is your daily profit and loss. Most days are near zero.



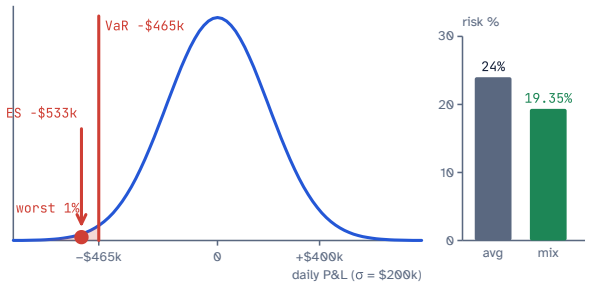
Step 2. The red sliver on the left is the worst 1% of days.



Step 3. VaR is where that sliver starts: a loss of $465k.



Step 4. ES is the average of everything inside the sliver: $533k. Further out than VaR.



Step 5. Diversification: mixing two assets gives 19.35% risk, less than the 24% average.

What it means

- **Volatility (σ):** the standard deviation of returns. Bigger σ = wider swings.
- **Diversification:** two assets that don't move together have a portfolio σ *below* the average of their σs. Correlation ρ controls how much.
- **VaR at 99%:** the loss you will not exceed on 99 out of 100 days. For a normal distribution it is 2.326 × σ.
- **Expected Shortfall (ES):** the average loss on the worst 1% of days. Always bigger than VaR, and it sees how fat the tail is. Regulators now prefer ES.

The formulas

```
Portfolio variance: σp² = w1²σ1² + w2²σ2² + 2·w1·w2·ρ·σ1·σ2
Normal VaR:  VaR = z × σ × Value      (z = 1.645 at 95%, 2.326 at 99%)
Normal ES:   ES  = σ × Value × φ(z) / (1 - confidence)
```

w = weights, ρ = correlation, φ = the normal bell-curve height.

Worked example

Question. A $10m portfolio has a daily σ of 2%. Find the 99% 1-day VaR and ES. Then: 60/40 in two assets with σ = 20% and 30%, ρ = 0.3. Portfolio σ?

1. Daily $σ = 0.02 × 10m = `$200,000`
2. VaR = 2.326 × 200,000 = `$465,000`
3. ES = 200,000 × φ(2.326)/0.01 = 200,000 × 2.665 = `$533,000`
4. $σp^2 = 0.36(0.04) + 0.16(0.09) + 2(0.6)(0.4)(0.3)(0.2)(0.3) =$ `0.03744`, σp = `19.35%`, below the 24% weighted average.

Answer: VaR $465k, ES $533k; portfolio σ 19.35%

Calculator example

Inputs: Portfolio value ($) = 10000000, Daily volatility (%) = 2, Confidence (95 or 99) = 99, Days = 1

```
$σ over 1 day(s) = 200,000.00
VaR = 2.3263 × 200,000.00 = $465,260.00
ES = 200,000.00 × φ(2.3263)/0.01 = $533,102.21
```

Common traps

- Calling VaR the worst possible loss. It is a threshold, not a maximum.
- Scaling 1-day VaR to 10 days by 10 instead of √10.
- Forgetting the 2·w1·w2 in the variance formula.

Where to read more

- `Risk/01 Course Information/FRE6123_INFO_01_Syllabus_Fall_2026.docx (RMN1)`
- `Risk lecture decks RMN1-2 are not in your folder yet`

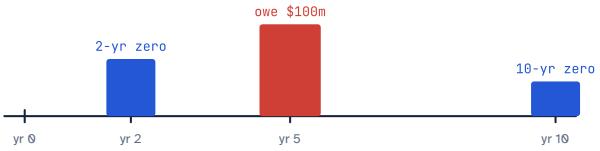
Immunization and asset-liability management

A pension fund or bank owes money in the future. Immunization picks assets whose value moves the same way as those debts when rates change, so a rate move does not hurt.

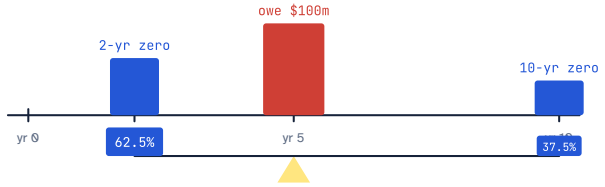
The sketch, step by step



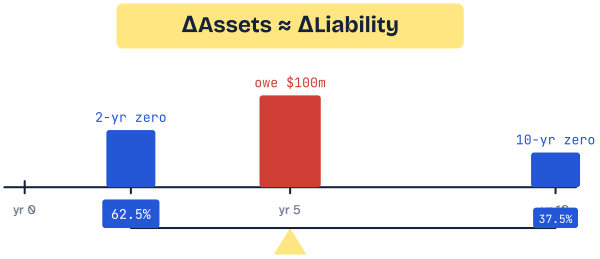
Step 1. A timeline. You owe $100m in present-value terms at year 5.



Step 2. You can only buy bonds that pay at year 2 and year 10.



Step 3. Think of a seesaw balanced at year 5: 62.5% at year 2 and 37.5% at year 10 balance it.



Step 4. Now if rates move a little, assets and the liability move by about the same amount. You are immunized.

What it means

- **ALM (asset-liability management):** manage assets and liabilities together, not separately. The risk is the *gap* between them.
- **Duration matching:** make the PV of assets equal the PV of liabilities, and their durations equal. Then a small parallel rate shift changes both by about the same amount.
- **Convexity:** also have asset convexity ≥ liability convexity, so bigger moves help rather than hurt.
- **Rebalance:** durations drift as time passes and rates move, so immunization must be redone.
- **Key-rate duration:** the curve does not always shift in parallel. Key-rate durations measure exposure to each part of the curve (2y, 5y, 10y…) separately.
- **Hedge accounting** (RMN8): rules that let a company show a hedge and the item it hedges together in its accounts, so the P&L does not jump around.

The formulas

```
Match PV:        PV(assets) = PV(liabilities)
Match duration:  w1·D1 + w2·D2 = D_L,   w1 + w2 = 1
Surplus change:  ΔS ≈ -(A·D_A - L·D_L) × Δy
```

w = share of money in each bond, D = duration, A and L = PV of assets and liabilities.

Worked example

Question. You owe $100m (PV) in 5 years. You can buy 2-year and 10-year zero-coupon bonds. How much of each?

1. Zero-coupon bond duration = its maturity: D1 = 2, D2 = 10.
2. Solve w1 × 2 + w2 × 10 = 5 with w1 + w2 = 1.
3. w2 × 8 = 3 → w2 = `0.375`, w1 = `0.625`
4. Buy $62.5m of 2-year zeros and $37.5m of 10-year zeros (in PV). This "barbell" also has more convexity than the liability.

Answer: 62.5% in 2-year, 37.5% in 10-year

Calculator example

Inputs: Liability duration = 5, Bond 1 duration = 2, Bond 2 duration = 10, PV of liability ($m) = 100

```
w2 = (5 - 2) / (10 - 2) = 0.3750, w1 = 0.6250
Buy $62.50m of bond 1 and $37.50m of bond 2 (PV)
```

Common traps

- Matching maturities instead of durations.
- Forgetting to also match PV.
- Assuming the hedge lasts forever; durations drift and must be rebalanced.

Where to read more

- `Risk syllabus RMN2 and RMN8 (duration, immunization, ALM, hedge accounting)`
- `Derivatives/03 Lectures/FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx`

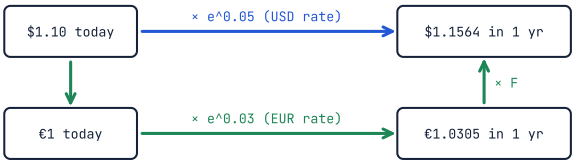
Currency and commodity risk

Exchange-rate forwards follow interest rate parity: the currency with the higher interest rate trades at a discount forward. Commodity forwards add storage costs and a convenience yield.

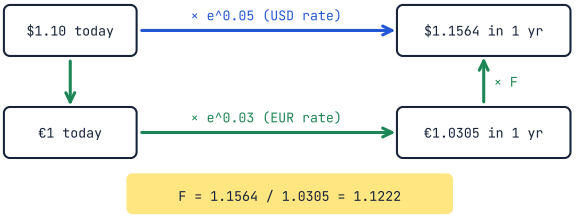
The sketch, step by step



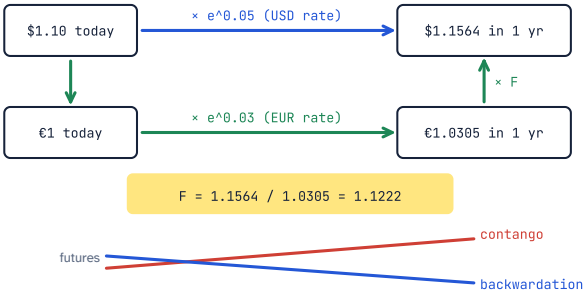
Step 1. Path 1: keep $1.10 in dollars at 5% for a year. You end with 1.10 × e^0.05.



Step 2. Path 2: change it to €1, earn 3% in euros, and convert back at the forward rate F.



Step 3. Both paths must give the same dollars, or there is free money. So F = 1.10 × e^0.02 = 1.1222.



Step 4. Commodities: storage pushes forward prices up (contango); a convenience yield pulls them down (backwardation).

What it means

- **FX quote:** EUR/USD = 1.10 means 1 euro costs 1.10 dollars.
- **Covered interest rate parity:** you cannot earn free money by borrowing in one currency, converting, investing in the other, and locking in the rate back. So $F = S \cdot e^{(r - r_f)T}$.
- **FX exposure:** a US firm with euro revenue loses if the euro falls. Hedge by selling euros forward.
- **Commodities:** holding oil costs storage (u). But having it on hand also has a benefit, the **convenience yield** (y). $F = S \cdot e^{(r + u - y)T}$.
- **Contango**: F > S (storage dominates). **Backwardation**: F < S (high convenience yield, e.g. a shortage).
- **Equity risk** (also RMN3): value a stock with dividends or factors (CAPM, multi-factor).

The formulas

```
FX forward:        F = S × e^((r - r_f)·T)
Commodity forward: F = S × e^((r + u - y)·T)
```

S = spot price (in domestic currency per unit), r = domestic rate, r_f = foreign rate, u = storage cost rate, y = convenience yield.

Worked example

Question. EUR/USD spot is 1.10. USD rate 5%, EUR rate 3% (continuous). What is the 1-year forward? A US exporter will receive €1m in a year. What can it lock in?

1. F = 1.10 × e^((0.05 − 0.03) × 1) = 1.10 × e^0.02
2. e^0.02 = 1.0202 → F = `1.1222` USD per EUR
3. The euro is at a premium because its interest rate is lower.
4. Sell €1m forward: locks in `$1,122,200` no matter where the euro goes.

Answer: F = 1.1222; locks in $1.1222m

Calculator example

Inputs: Spot (USD per EUR) = 1.1, Domestic rate r (%) = 5, Foreign rate r_f (%) = 3, Years = 1, Foreign amount = 1000000

```
Forward = 1.1 × e^(0.0200 × 1) = 1.1222
Locked-in amount = 1,000,000 × 1.1222 = 1,122,221.47
```

Common traps

- Flipping r and r_f (check: the lower-rate currency is at a forward premium).
- Mixing up which currency the quote is "per".
- Leaving out convenience yield when the futures curve is downward sloping.

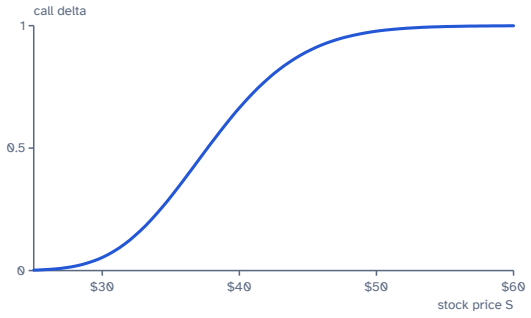
Where to read more

- Risk syllabus RMN3 (FX exposures and parity, equity, commodity carry and basis)
- Derivatives/03 Lectures/FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx

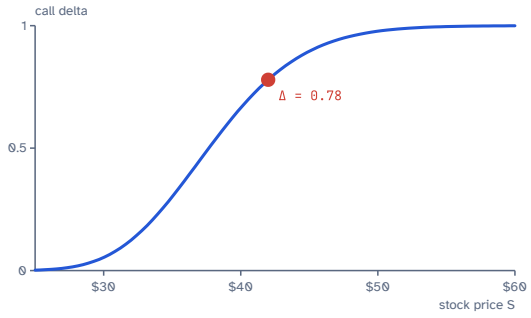
Greeks and delta hedging

A trader who sells options removes most of the risk by holding delta shares of the stock, then adjusting as delta changes. Gamma says how often you need to adjust.

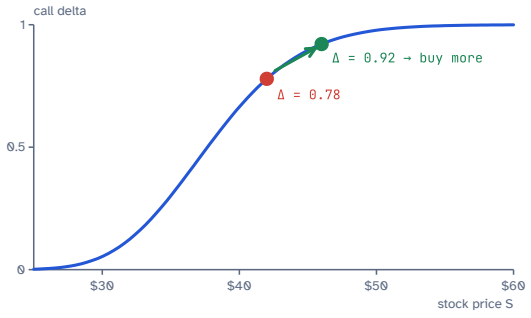
The sketch, step by step



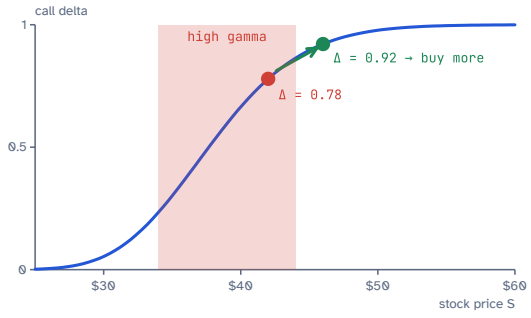
Step 1. Delta of a call (K = 40) against the stock price. It goes from 0 to 1.



Step 2. At S = 42 delta is about 0.78. Short 1 call → buy 0.78 shares.



Step 3. The stock rises to 46. Delta has climbed (that climb is gamma). You must buy more shares.



Step 4. The steepest part of the curve is near the strike: that is where gamma is highest and you rebalance most.

What it means

- **Delta-neutral:** total delta of the book = 0, so small stock moves do not change its value.
- Sold calls have negative delta, so you **buy** shares to hedge. Sold puts have positive delta, so you **sell** shares.
- **Gamma:** delta changes as the stock moves. New delta ≈ old delta + Γ × ΔS. High gamma = rebalance often.
- A short-option, delta-hedged book has negative gamma: it loses on big moves, and earns theta (time decay) on quiet days.
- **Gamma/vega hedging:** stock has zero gamma and vega, so you need other options to hedge those.
- **Hedging error:** you only rebalance at discrete times and volatility is not what you assumed, so the hedge is never perfect.

The formulas

```
Shares to hold = -(option position) × Δ
New Δ ≈ Δ + Γ × ΔS
P&L of hedged short option ≈ θ·Δt - ½·Γ·(ΔS)²   (per option, sign for a short)
```

Δ = delta, Γ = gamma, θ = theta (here per unit time, positive for a short option).

Worked example

Question. You sold 1,000 calls with delta 0.60 and gamma 0.04. How many shares to buy? The stock then rises $1.25. What do you do?

1. Option delta = −1,000 × 0.60 = −600 → buy `600 shares`
2. New delta ≈ 0.60 + 0.04 × 1.25 = `0.65`
3. You now need 1,000 × 0.65 = 650 shares
4. Buy `50 more`. (Notice: you buy after a rise and sell after a fall. That is how negative gamma costs money.)

Answer: Buy 600 shares, then 50 more

Calculator example

Inputs: Calls sold = 1000, Delta = 0.6, Gamma = 0.04, Stock move ($) = 1.25

```
Start: buy 600 shares
New delta ≈ 0.6 + 0.04 × 1.25 = 0.6500
Adjust: buy 50 shares
```

Common traps

- Selling shares to hedge a short call (it is buy).
- Thinking delta-neutral means no risk; gamma and vega remain.
- Forgetting the contract multiplier (1 contract often = 100 shares).

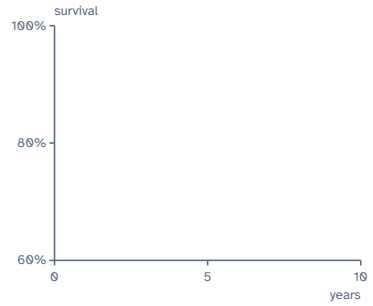
Where to read more

- Hull 8e, Chapter 18 (The Greek letters)
- Risk syllabus RMN4 (Greeks, dynamic hedging)

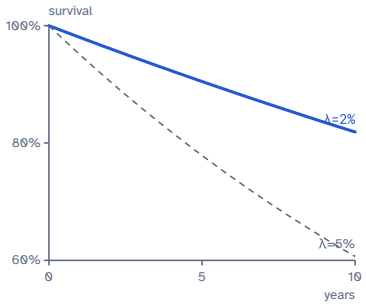
Credit risk: default, loss and spreads

Expected credit loss is chance of default × share you lose × amount at stake. A credit spread is the extra yield lenders demand to cover that.

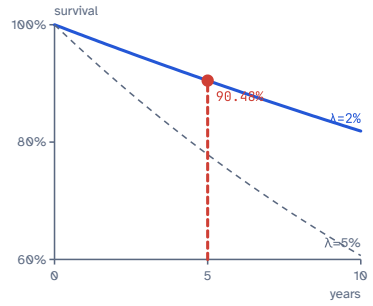
The sketch, step by step



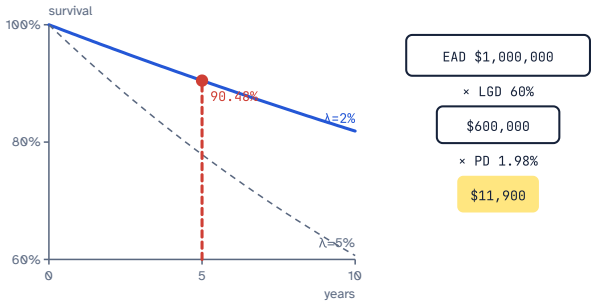
Step 1. The chance a borrower is still alive (not defaulted) over time.



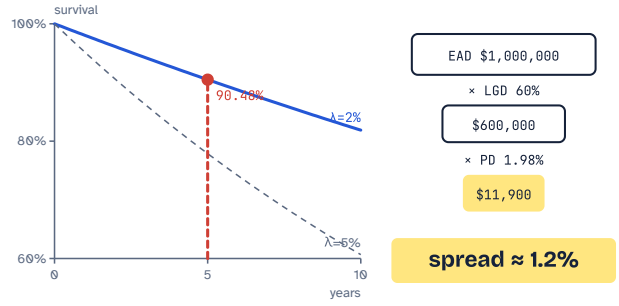
Step 2. With λ = 2% a year, survival slides slowly down: e^(−0.02t).



Step 3. At 5 years, 90.48% survive. So 9.52% have defaulted.



Step 4. Expected loss stacks three things: $1m at stake × 60% lost × 1.98% chance = $11.9k.



Step 5. The lender wants about λ × LGD = 1.2% extra yield to cover this.

What it means

- **PD** (probability of default), **LGD** (loss given default = 1 – recovery rate), **EAD** (exposure at default).
- **Hazard rate λ:** the default "speed". Survival to time t is $e^{-\lambda t}$; PD by t is $1 - e^{-\lambda t}$.
- **Credit spread ≈ λ × LGD.** A CDS (credit default swap) is insurance on a bond, and its spread works out to about the same number.
- **P vs Q:** real-world (P) default probabilities come from history. Risk-neutral (Q) ones come from prices and are higher, because they include a risk premium.
- **Expected loss** is a cost you price in. **Unexpected loss** is what capital must absorb.

The formulas

```
Survival:      S(t) = e^(-λt)        PD(t) = 1 - e^(-λt)
Expected loss: EL = PD × LGD × EAD
Spread:        s ≈ λ × LGD   →   λ ≈ s / (1 - R)
```

λ = hazard rate per year, R = recovery rate.

Worked example

Question. A borrower has λ = 2% a year. LGD = 60%. Exposure $1m. Find 5-year survival, 5-year PD, 1-year expected loss and the spread.

1. Survival 5y = e^(−0.02 × 5) = e^(−0.1) = `90.48%`
2. PD 5y = 1 − 0.9048 = `9.52%`
3. 1-year PD ≈ 1 − e^(−0.02) = 1.98%, EL ≈ 0.0198 × 0.6 × 1,000,000 = `$11,900`
4. Spread ≈ 0.02 × 0.6 = `1.2%` = 120 basis points

Answer: Survival 90.48%, PD 9.52%, EL ≈ $11.9k, spread ≈ 120 bp

Common traps

- Using recovery rate where LGD is needed (or the reverse).
- Adding PDs across years instead of using survival.
- Mixing P and Q probabilities in the same calculation.

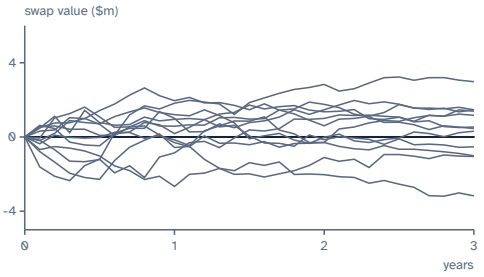
Where to read more

- `Valuation/03 Lectures/FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf`
- `Risk syllabus RMN5-6 (PD, LGD, CDS, CVA)`
- `Hull 8e, Chapter 23 (Credit risk)`

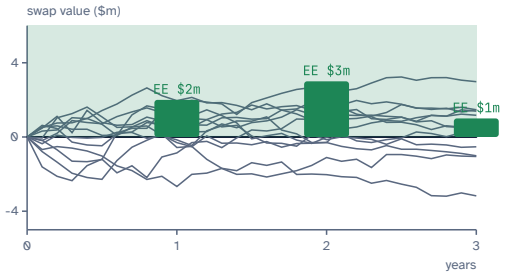
Counterparty risk and CVA

If the other side of a derivative defaults while it owes you money, you lose. CVA is the price of that risk, built from exposure, default probability and loss given default.

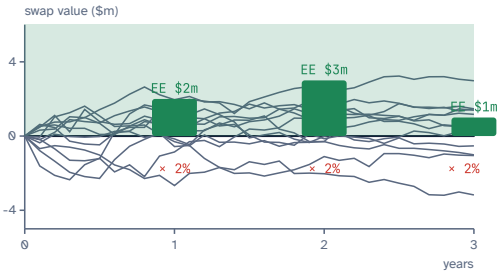
The sketch, step by step



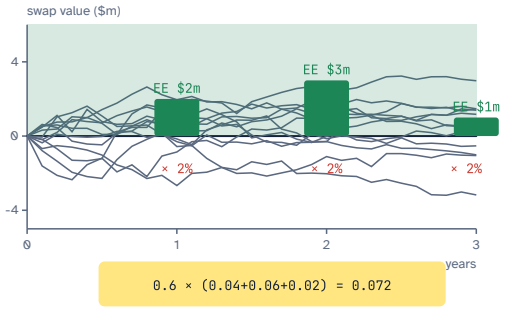
Step 1. Possible future values of a swap. Some paths end up in your favour (above 0), some against you.



Step 2. Only the part above zero is at risk. Averaging that gives the expected exposure: a hump over time.



Step 3. Multiply each year's exposure by the chance of default in that year.



Step 4. Add them up and multiply by LGD: CVA ≈ $72k.

What it means

- **Exposure:** only positive value matters. If the trade is worth +5 to you and they default, you lose up to 5. If it is worth −5, you still owe them.
- **Expected exposure EE(t):** the average of max(V, 0) across simulated futures. For a swap it rises then falls (a hump), because fewer payments remain later.
- **PFE (potential future exposure):** a high percentile (e.g. 95%) of exposure. Used for limits.
- **CVA (credit valuation adjustment):** expected loss from counterparty default, subtracted from the trade value.
- **Reducing it:** netting, collateral (margin), and central clearing. xVA adds related adjustments (DVA, FVA, KVA).
- **Migration:** a counterparty can be downgraded before defaulting, which raises CVA too.

The formulas

```
Exposure:  E(t) = max(V(t), 0)
CVA ≈ LGD × Σ EE(t_i) × PD(t_{i-1}, t_i) × DF(t_i)
```

PD(t_{i−1}, t_i) = probability of default in that period, DF = discount factor.

Worked example

Question. A 3-year swap has expected exposure $2m, $3m, $1m in years 1, 2, 3. The counterparty has a 2% chance of defaulting in each year; LGD 60%. Ignore discounting. CVA?

1. Year 1: 2 × 0.02 = `0.04`
2. Year 2: 3 × 0.02 = `0.06`; Year 3: 1 × 0.02 = `0.02`
3. Sum = 0.12, times LGD 0.6 = `0.072`
4. CVA ≈ $72,000. The trade is worth $72k less than if the counterparty were risk-free.

Answer: CVA ≈ $72,000

Common traps

- Using V instead of max(V, 0).
- Using cumulative PD for every year instead of the PD within each period.
- Forgetting that collateral and netting cut exposure.

Where to read more

- Risk syllabus RMN5-6 (CVA, exposure profiles, xVA)
- Hull 8e, Chapter 24 (Credit risk)

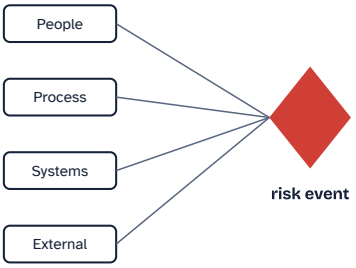
Operational risk and controls

Operational risk is loss from failed people, processes, systems or outside events: a fat-finger trade, a hack, a payment sent twice, fraud.

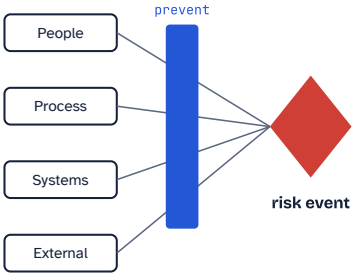
The sketch, step by step



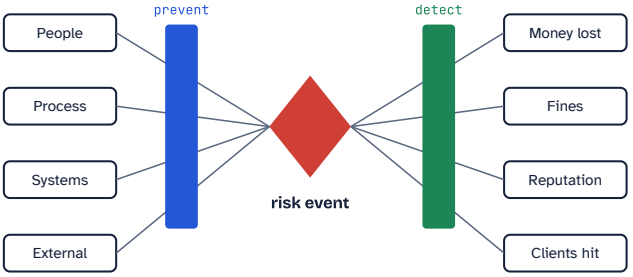
Step 1. A bow-tie: the risk event sits in the middle (e.g. a payment sent twice).



Step 2. On the left are the causes: people, process, systems, external events.



Step 3. Preventive controls sit between causes and the event. They try to stop it.



Step 4. On the right are the consequences. Detective controls and recovery plans limit the damage.

What it means

- **Basel definition:** loss from inadequate or failed internal processes, people and systems, or from external events. Includes legal risk.
- **Frequency × severity:** many small losses (errors) and rare huge ones (fraud, rogue trading). The rare ones dominate the tail.
- **Controls:** preventive (stop it happening: approvals, access limits) and detective (catch it: reconciliations, alerts).
- **KRIs (key risk indicators):** early-warning numbers, e.g. failed trades per day, staff turnover, overdue reconciliations.
- **Three lines of defense:** 1) the business owns the risk, 2) risk & compliance oversee, 3) internal audit checks independently.
- **Payments and settlement risk:** paying before you get paid (Herstatt risk). Fixed by delivery-versus-payment and CLS for FX.
- **Technology/cyber risk** and model risk sit here too.

The formulas

```
Expected annual loss ≈ expected frequency × expected severity
Residual risk = inherent risk - effect of controls
```

Op-risk capital looks at the tail of the loss distribution, not just the average.

Worked example

Question. A desk has about 12 settlement errors a year, each costing $50,000 on average. A new reconciliation control would catch 75% of them and costs $200,000 a year. Is it worth it (just on expected loss)?

1. Expected loss now: 12 × 50,000 = `$600,000` a year
2. Losses avoided: 75% × 600,000 = `$450,000`
3. Net benefit: 450,000 − 200,000 = `$250,000` a year
4. Yes, and it also cuts tail risk (fewer chances of one giant error).

Answer: Worth it: saves about $250,000 a year

Common traps

- Looking only at average losses and ignoring rare severe events.
- Confusing a control (an action) with a KRI (a measurement).
- Treating operational risk as only IT risk.

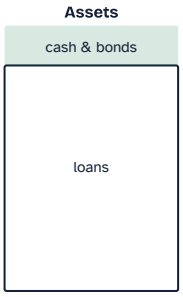
Where to read more

- Risk syllabus RMN7 (operational risk, payments, controls, KRIs, technology risk)

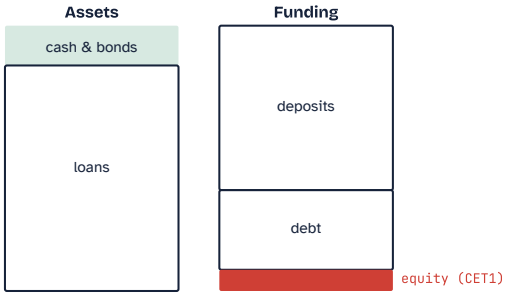
Bank capital and liquidity rules

Banks must keep enough capital (their own money) to absorb losses and enough liquid assets to survive a month of panic withdrawals. Basel III sets the minimums.

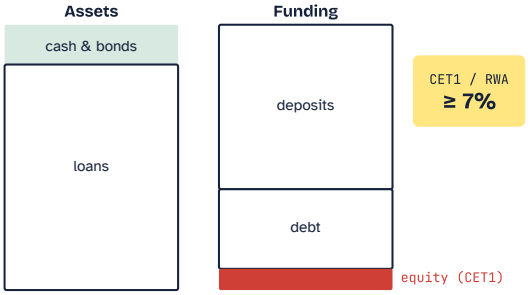
The sketch, step by step



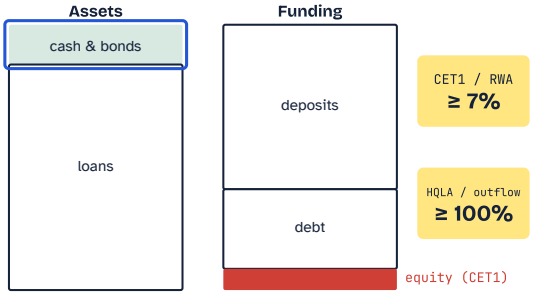
Step 1. A bank balance sheet: assets on the left, how they are funded on the right.



Step 2. Most funding is deposits and debt. A thin slice is equity: that is capital.



Step 3. Weight the assets by risk to get RWA. Capital must be at least 7% of that (with the buffer).



Step 4. Liquidity is a separate test: enough liquid assets to cover 30 days of stressed outflows.

What it means

- **Capital:** equity that can absorb losses. CET1 (common equity tier 1) is the best quality.
- **Risk-weighted assets (RWA):** each asset is weighted by riskiness (cash ~0%, mortgages lower, corporate loans ~100%). Includes market and operational risk too.
- **Ratios (Basel III minimums):** CET1 ≥ 4.5% of RWA, Tier 1 ≥ 6%, total capital ≥ 8%, plus a 2.5% conservation buffer (and extra for big banks).
- **Leverage ratio:** Tier 1 / total exposure (not risk weighted) ≥ 3%. A backstop if risk weights are wrong.
- **LCR (liquidity coverage ratio):** high-quality liquid assets / net cash outflows over 30 days of stress ≥ 100%.
- **NSFR (net stable funding ratio):** stable funding / required stable funding ≥ 100% over one year.

The formulas

```
CET1 ratio = CET1 / RWA              (≥ 4.5%, ≥ 7% with buffer)
Leverage   = Tier 1 / Total exposure  (≥ 3%)
LCR        = HQLA / 30-day net outflows (≥ 100%)
```

HQLA = high-quality liquid assets (cash, central bank reserves, government bonds).

Worked example

Question. A bank has CET1 $60bn, RWA $800bn, total exposure $1,500bn, HQLA $120bn and stressed 30-day outflows of $100bn. Assume Tier 1 = CET1. Does it pass?

1. CET1 ratio = 60 / 800 = `7.5%` ≥ 7% ✓
2. Leverage = 60 / 1,500 = `4.0%` ≥ 3% ✓
3. LCR = 120 / 100 = `120%` ≥ 100% ✓
4. It passes all three, but only 0.5% above the CET1 buffer: a $4bn loss would breach it.

Answer: 7.5% CET1, 4% leverage, 120% LCR: passes

Calculator example

Inputs: CET1 ($bn) = 60, RWA ($bn) = 800, Total exposure ($bn) = 1500, HQLA ($bn) = 120, 30-day outflows ($bn) = 100

```
CET1 ratio = 7.50% ✓ (need 7%)
Leverage = 4.00% ✓ (need 3%)
LCR = 120% ✓ (need 100%)
```

Common traps

- Dividing capital by total assets for the CET1 ratio (use RWA).
- Thinking capital is cash. Capital is how the bank is funded, not what it holds.
- Mixing up liquidity (cash to pay today) and solvency (enough capital overall).

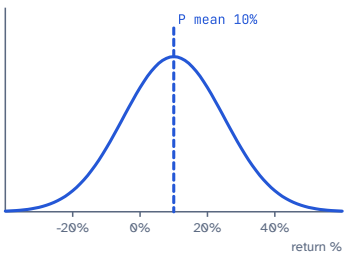
Where to read more

- Risk syllabus RMN9 (bank balance sheet, capital, leverage, liquidity)
- BIS Basel Framework (see "Watch online")

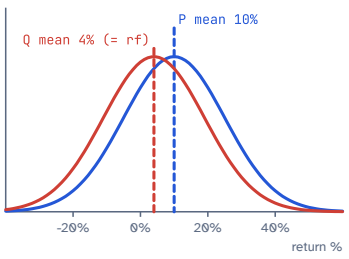
P vs Q, risk drivers and stress tests

P is the real world (what actually tends to happen). Q is the risk-neutral world used for pricing. Risk managers measure losses in P, price in Q, and stress-test both.

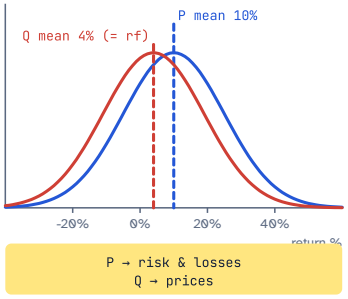
The sketch, step by step



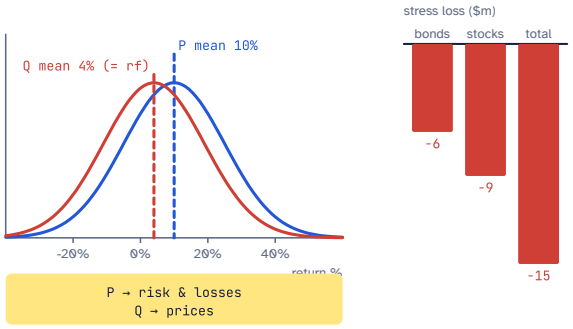
Step 1. P: the real-world spread of outcomes for a risky asset. Its average is above the risk-free rate.



Step 2. Q: the same outcomes reweighted so the average is the risk-free rate. More weight on bad outcomes.



Step 3. Use P for risk (VaR, expected loss). Use Q for prices.



Step 4. Stress test: shock each driver and add the hits. Bonds –$6m, equities –$9m, total –$15m.

What it means

- **Q (risk-neutral):** probabilities that make every asset earn the risk-free rate. Used for pricing (binomial trees, Black-Scholes, CDS). Not a forecast.
- **P (physical/real world):** probabilities from history and judgement. Used for VaR, expected losses, capital.
- Q puts more weight on bad outcomes because investors dislike them. That is why Q default probabilities from spreads are higher than historical ones.
- **Risk drivers:** the few market factors that move a portfolio (rates, spreads, equity index, FX, vol). Map each position to them.
- **Aggregation:** combine risks across desks using correlations; diversification lowers the total, but correlations jump in a crisis.
- **Ex-ante P&L / stress tests:** apply a scenario (e.g. rates +200bp, equities –30%, spreads +150bp) and reprice everything to see the loss before it happens.

The formulas

```
Q default intensity from spread:  λQ ≈ s / LGD
Stress loss ≈ Σ (sensitivity to driver × shock)    (or full revaluation)
Bond:  ΔV ≈ -V × D × Δy
```

Use P for "how likely / how much could we lose", Q for "what is it worth".

Worked example

Question. A bond portfolio worth $50m (duration 6) and an equity portfolio worth $30m. Stress: rates +2%, equities −30%. Loss? Also: a spread of 150bp with 60% LGD implies what Q default rate, if history says 1%?

1. Bonds: −50 × 6 × 0.02 = `-$6m`
2. Equities: −30% × 30 = `-$9m`
3. Total stress loss ≈ `$15m` (19% of $80m)
4. λQ ≈ 0.015 / 0.6 = `2.5%` a year vs 1% in P. The gap is the credit risk premium.

Answer: Stress loss ≈ $15m; Q default rate 2.5% vs P 1%

Common traps

- Using Q probabilities to forecast real default rates.
- Adding stress losses that cannot happen together, or ignoring ones that do.
- Trusting normal-times correlations in a stress scenario.

Where to read more

- `Risk syllabus RMNA (P vs Q, drivers, aggregation, ex-ante P&L, stress)`
- `Valuation notes: instructor-linked explanation of risk neutrality`

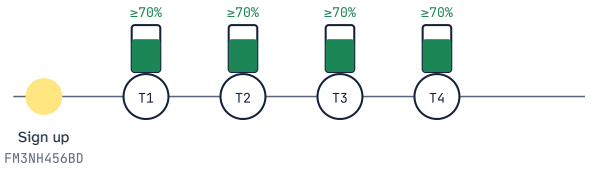
Bloomberg: passing the four tests

The Bloomberg course is pass/fail on four tests: at least 70% on each one, done by December 14, under class code FM3NH456BD.

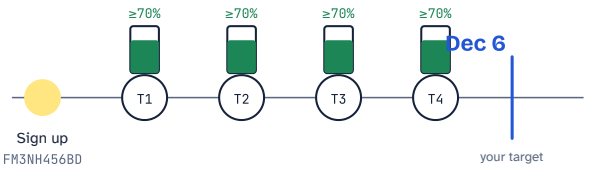
The sketch, step by step



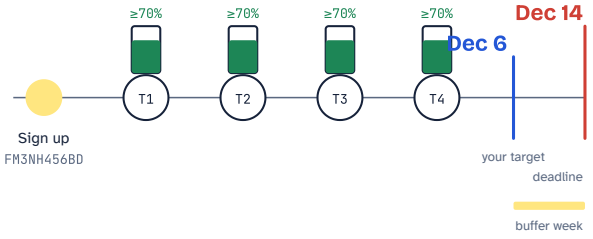
Step 1. Step 1: sign up and enter class code FM3NH456BD.



Step 2. Then four tests along the way. Each needs at least 70%.



Step 3. Your own target finish line is December 6.



Step 4. The real deadline is December 14. The week in between is your safety buffer.

What it means

- Sign up with the access guide in your folder and enter class code `FM3NH456BD` so the instructor can see your scores.
- Watching videos or getting a certificate is **not** enough. All four test scores must be 70% or more.
- Do the lessons actively: pause, note unfamiliar terms, and link them to what you learned here (yields and duration, futures, options, credit).
- Aim to finish by your own target of **December 6**, leaving a week of buffer before the real deadline of **December 14**.
- After each test, check the score shows on your class record.

The formulas

```
Pass = all four tests ≥ 70%
Deadline: Mon Dec 14, 2026 (target Dec 6)
```

Bloomberg lessons and tests are online, not in your folder.

Worked example

Question. Plan: you have 30-minute Bloomberg slots twice a week from Nov 2. How do you fit four tests before Dec 6?

1. Week of Nov 2: register, confirm the class code, start the first module.

2. Weeks of Nov 9 and Nov 16: finish modules and take tests 1 and 2.
3. Weeks of Nov 23 and Nov 30: tests 3 and 4 (book a longer session if a test needs it).
4. Retake any test under 70%, then check all four scores appear.

Answer: Four tests done by Dec 6, one week of buffer

Common traps

- Not entering the class code, so the instructor never sees your scores.
- Assuming a certificate means you passed.
- Leaving all four tests to the final week.

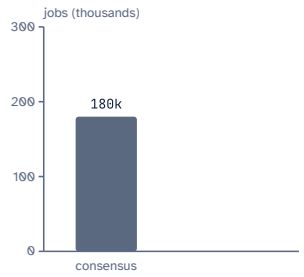
Where to read more

- `Bloomberg/01 Instructions and Requirements/FRE5500_NOTES_01_Bloomberg_Signup_and_BMC_Access_Summer_2025.pdf`
- `Bloomberg/01 Instructions and Requirements/FRE5500_NOTES_02_Requirements.md`

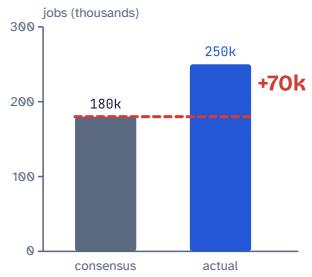
Economic indicators

Markets move on surprises, not on the news itself: what matters is the actual number minus what economists expected.

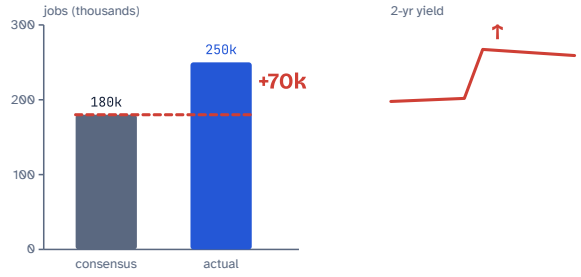
The sketch, step by step



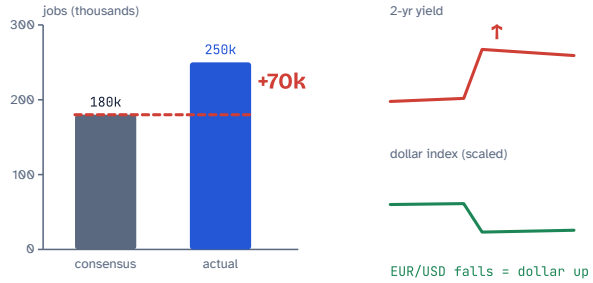
Step 1. The economists' consensus forecast: +180k jobs.



Step 2. The release: +250k. A +70k surprise.



Step 3. The market reacts within seconds: 2-year yield jumps up.



Step 4. And the dollar rises against other currencies.

What it means

- **GDP:** total output. Growth rate is the headline. Released quarterly.
- **Inflation (CPI, PCE):** how fast prices rise. Central banks target about 2%.
- **Jobs (nonfarm payrolls, unemployment rate):** monthly; one of the biggest market movers in the US.
- **Surveys (PMI, consumer confidence):** early signals. PMI above 50 = expanding.
- **Leading vs lagging:** leading indicators (PMI, building permits) turn before the economy; lagging ones (unemployment) after.
- **Central bank link:** strong growth or high inflation → expect higher policy rates → bond yields up, currency up, often stocks down.
- On the terminal, the economic calendar shows the release time, the consensus (survey) and the actual.

The formulas

```
Surprise = Actual - Consensus
Positive surprise (strong data) → yields ↑, currency ↑
Negative surprise → yields ↓, currency ↓
```

Typical reactions. Real markets can react differently if the details disagree with the headline.

Worked example

Question. Consensus for US payrolls is +180k. The actual is +250k. Unemployment is unchanged. What do you expect for Treasury yields and the dollar?

1. Surprise = 250k − 180k = `+70k`, a strong beat
2. Stronger jobs → the Fed is less likely to cut rates
3. Short-term Treasury yields `rise` (prices fall)
4. The dollar `strengthens`, as higher US rates attract money

Answer: Yields up, dollar up

Common traps

- Reacting to the level instead of the surprise.
- Mixing up month-over-month and year-over-year inflation.
- Forgetting that a bond price falls when its yield rises.

Where to read more

- `Bloomberg/01 Instructions and Requirements/FRE5500_NOTES_02_Requirements.md`
- `Bloomberg Market Concepts: Economic Indicators module`

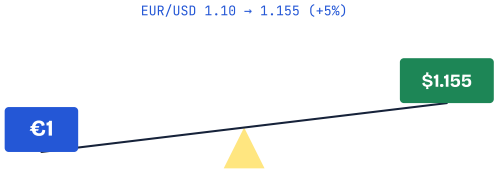
Currencies

An exchange rate is just the price of one currency in another. Interest rates, inflation, growth and risk appetite push it around.

The sketch, step by step



Step 1. A seesaw: EUR on the left, USD on the right. At 1.10, one euro balances 1.10 dollars.



Step 2. The ECB raises rates; money flows into euros. The euro side gets heavier: EUR/USD rises to 1.155.



Step 3. Cross rates: if you know EUR/USD and USD/JPY you can get EUR/JPY by multiplying.

What it means

- **Quote:** EUR/USD = 1.10 means 1 EUR (the base) = 1.10 USD (the quote currency). If it rises to 1.12, the euro got stronger.
- **Pips:** the 4th decimal (0.0001). Spread = ask − bid.
- **Interest rate differentials:** higher rates attract money → currency tends to strengthen (spot). Forwards follow interest rate parity (see the Risk class on currencies).
- **Carry trade:** borrow in a low-rate currency, invest in a high-rate one. Earns the gap, until a sudden reversal.
- **PPP (purchasing power parity):** in the long run, the same basket should cost the same everywhere; high-inflation currencies weaken.
- **Safe havens** (USD, JPY, CHF) tend to rise in a crisis.
- **Who cares:** exporters, importers, investors holding foreign assets, central banks.

The formulas

```
Amount in quote currency = amount in base × rate
Cross rate: EUR/JPY = EUR/USD × USD/JPY
% change of base = new rate / old rate - 1
```

Always write the pair as BASE/QUOTE and keep units: QUOTE per 1 BASE.

Worked example

Question. EUR/USD = 1.10 and USD/JPY = 150. (a) How many dollars is €2m? (b) What is EUR/JPY? (c) EUR/USD moves to 1.155. How much did the euro gain?

1. (a) 2,000,000 × 1.10 = `$2,200,000`
2. (b) EUR/JPY = 1.10 × 150 = `165` yen per euro
3. (c) 1.155 / 1.10 − 1 = `+5%`
4. So a US investor holding euro assets gained 5% from the currency alone.

Answer: $2.2m; EUR/JPY 165; euro +5%

Calculator example

Inputs: Amount of base currency = 2000000, Rate (quote per base) = 1.1, New rate = 1.155

```
Converted = 2,000,000 × 1.1 = 2,200,000.00
Base currency change = 1.155 / 1.1 - 1 = 5.00%
Same amount at new rate = 2,310,000.00
```

Common traps

- Multiplying when you should divide (check units: USD per EUR × EUR = USD).
- Reading a fall in EUR/USD as the dollar weakening (it is the dollar strengthening).
- Forgetting the currency effect on foreign investments.

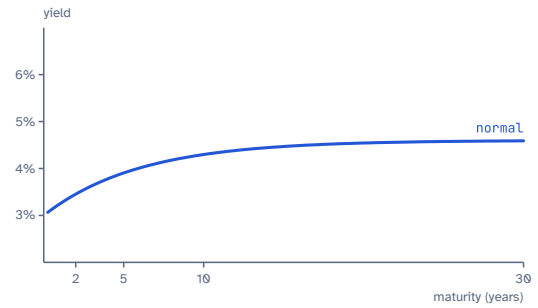
Where to read more

- Bloomberg Market Concepts: Currencies module

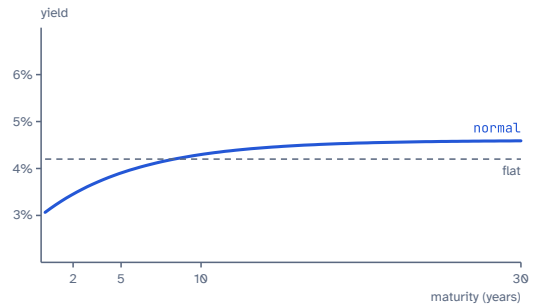
Fixed income and the yield curve

A bond's price and yield move in opposite directions. The yield curve (yields by maturity) shows what the market expects for rates and growth.

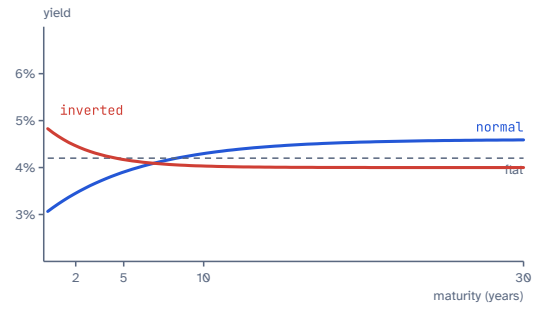
The sketch, step by step



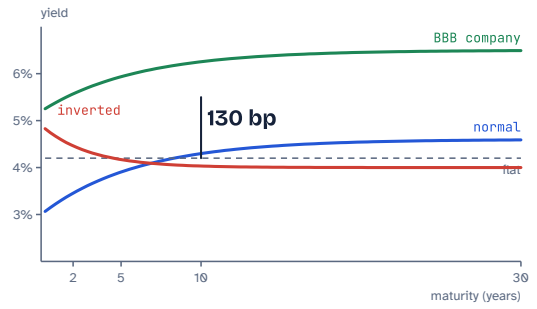
Step 1. A normal yield curve: you get paid more to lend for longer.



Step 2. A flat curve: all maturities pay about the same.



Step 3. An inverted curve: short rates above long rates. Markets expect cuts, often a recession warning.



Step 4. Add a company's curve above the Treasury curve. The gap is the credit spread: 130 bp at 10 years.

What it means

- **Price vs yield:** rates up → existing bonds with lower coupons are worth less → price down.
- **Yield curve shapes:** normal (upward: longer = higher yield), flat, inverted (short rates above long, often seen before recessions).
- **Central banks** set short rates; long rates reflect expected future short rates plus a term premium.
- **Credit spread:** corporate bond yield − Treasury yield of the same maturity. Wider spread = more perceived default risk.
- **Ratings:** AAA … BBB− is investment grade; BB+ and below is high yield ("junk").
- **Duration** (see the Derivatives class) tells you price sensitivity: duration 7 → about −7% for +1% in yield.

The formulas

```
Credit spread = corporate yield - Treasury yield
Curve slope   = 10-yr yield - 2-yr yield
% price change ≈ -Duration × Δyield
```

1 basis point (bp) = 0.01%.

Worked example

Question. 10-year Treasury yields 4.20%. A BBB company's 10-year bond yields 5.50%. The 2-year Treasury yields 4.60%. Spread? Curve shape? If the corporate bond (duration 8) sees its yield rise 0.25%, price change?

1. Credit spread = 5.50 − 4.20 = `1.30% = 130 bp`
2. Slope = 4.20 − 4.60 = `-40 bp` → the curve is inverted
3. Price change ≈ −8 × 0.25% = `-2%`
4. Inverted curve: markets expect rate cuts ahead, often linked with slowing growth.

Answer: Spread 130 bp; inverted curve; price −2%

Common traps

- Saying a bond price rises when yields rise.
- Comparing yields of different maturities when computing a spread.
- Writing 130 bp as 13%.

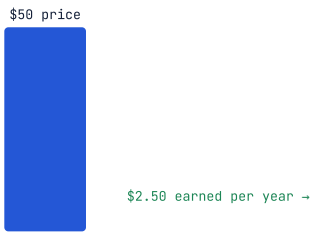
Where to read more

- `Bloomberg Market Concepts: Fixed Income module`
- `Valuation/03 Lectures/FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf`

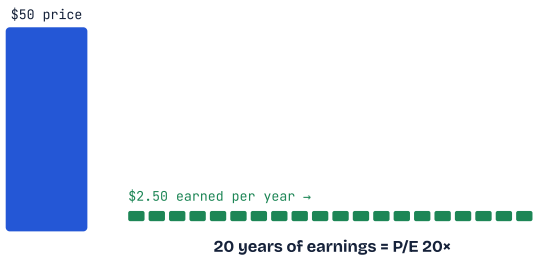
Equities and valuation multiples

A share is a slice of a company's future profits. Multiples like P/E are a quick way to compare how "expensive" companies are.

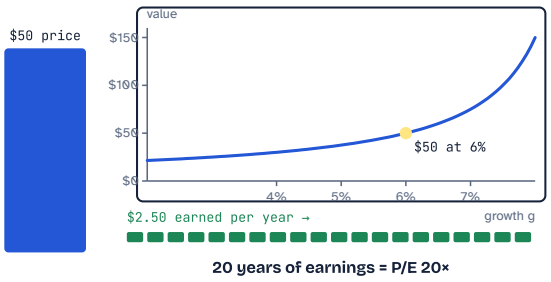
The sketch, step by step



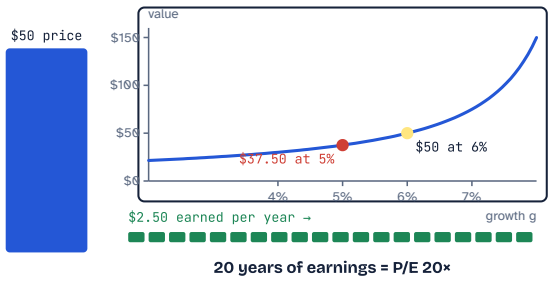
Step 1. A $50 share. Each year it earns $2.50 per share.



Step 2. So you pay 20 years of today's profit: P/E = 20×.



Step 3. The Gordon model values the growing stream of dividends: 1.50 / (9% − 6%) = $50.



Step 4. Small changes in growth move the value a lot: at 5% growth the value drops to $37.50.

What it means

- **P/E = price / earnings per share.** How many dollars you pay for $1 of yearly profit. Higher can mean faster expected growth, or overpricing.
- **EV/EBITDA:** enterprise value (equity + debt − cash) over operating earnings. Compares firms with different debt levels.
- **Dividend yield** = dividend / price. **Market cap** = price × shares.
- **DCF (discounted cash flow):** the "proper" value: PV of future free cash flows (Valuation classes). A simple version is the Gordon model: $P = D_1/(r - g)$.
- **Indices** (S&P 500, etc.) track groups of stocks. **Beta** measures how much a stock moves with the index (Portfolio/CAPM class).
- **What moves stocks:** earnings surprises, guidance, interest rates (higher rates = lower PV of future profits), sector news.

The formulas

```
P/E = Price / EPS
EV = Market cap + Debt - Cash
Gordon growth: P0 = D1 / (r - g)
```

EPS = earnings per share, D1 = next year's dividend, r = required return, g = growth rate.

Worked example

Question. A stock is $50 with EPS $2.50 and next year's dividend $1.50. Required return 9%, dividends grow 6% a year. P/E? Gordon value? Is it cheap?

1. P/E = 50 / 2.50 = `20×`
2. Gordon value = 1.50 / (0.09 − 0.06) = 1.50 / 0.03 = `$50`
3. Model value = market price, so fairly priced under these assumptions.
4. If growth were only 5%: 1.50 / 0.04 = $37.50, so the price assumes 6% growth.

Answer: P/E 20×; Gordon value $50 (fairly priced)

Calculator example

Inputs: Price = 50, EPS = 2.5, Next dividend D1 = 1.5, Required return (%) = 9, Growth (%) = 6

```
P/E = 20.0×
Gordon value = 1.5 / (0.09 - 0.06) = $50.00
```

Common traps

- Comparing P/E across very different industries.
- Using Gordon when g ≥ r (the formula breaks).
- Forgetting to subtract cash and add debt for EV.

Where to read more

- `Bloomberg Market Concepts: Equities module`
- `Valuation/02 Books and Readings/FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf`

Watch and learn online

Free videos and courses to use alongside these classes. Coursera courses are free when you choose "audit".

Derivatives

- **Khan Academy: Finance and capital markets** (Free short videos, very beginner)
  https://www.khanacademy.org/economics-finance-domain/core-finance
  Watch the units on forwards, futures, options, put-call parity and swaps first. Best place to start if a class here feels too fast.
- **MIT OpenCourseWare 18.S096: Mathematics with Applications in Finance** (Free full lecture videos (MIT, 2013))
  https://ocw.mit.edu/courses/18-s096-topics-in-mathematics-with-applications-in-finance-fall-2013/
  Watch the lectures on stochastic processes, Itô calculus and Black-Scholes once you finish the Binomial and Black-Scholes classes here.
- **Columbia (Coursera): Introduction to Financial Engineering and Risk Management** (Free to audit, video course)
  https://www.coursera.org/learn/financial-engineering-intro
  Covers no-arbitrage, forwards, swaps, binomial trees and option pricing in the same order as your course.
- **Bionic Turtle (YouTube)** (Free short videos on Hull topics)
  https://www.youtube.com/@bionicturtle
  Search the channel for any Hull chapter title (e.g. "hedge ratio", "binomial", "greeks") for a 5–10 minute explanation.

Valuation

- **Your course: Week 1 recording (Prof. Shimko)** (Your own class video)
  https://www.youtube.com/watch?v=5t3T9z_fM4U
  Start here: it is your instructor explaining present value and the course setup.
- **Aswath Damodaran (NYU Stern): free valuation classes** (Free lecture videos, slides and spreadsheets)
  https://pages.stern.nyu.edu/~adamodar/New_Home_Page/home.htm
  His Valuation and Corporate Finance classes (also on his YouTube channel) cover DCF, discount rates, multiples and real options in plain language.
- **MIT OpenCourseWare 15.401: Finance Theory I (Andrew Lo)** (Free full lecture videos)
  https://ocw.mit.edu/courses/15-401-finance-theory-i-fall-2008/
  Present value, bonds, CAPM and capital budgeting: matches the Valuation classes almost one-to-one.
- **Yale (Coursera): Financial Markets with Robert Shiller** (Free to audit, video course)
  https://www.coursera.org/learn/financial-markets-global
  Good big-picture lectures on risk, portfolio theory, CAPM and how markets work.
- **Khan Academy: Finance and capital markets** (Free short videos, very beginner)
  https://www.khanacademy.org/economics-finance-domain/core-finance
  Units on interest and present value, bonds, CDOs and mortgage-backed securities.

Risk Management

- **Bionic Turtle (YouTube)** (Free FRM-style videos)
  https://www.youtube.com/@bionicturtle
  Best free match for this course: VaR vs expected shortfall, duration, credit risk, CVA, operational risk and Basel. Search the channel by topic.
- **TU Delft OpenCourseWare: Special VaRs and Expected Shortfall** (Free university video lecture)
  https://ocw.tudelft.nl/course-lectures/3-2-special-vars-expected-shortfall/
  A clear short lecture on VaR and ES with pictures.
- **Columbia (Coursera): Introduction to Financial Engineering and Risk Management** (Free to audit, video course)
  https://www.coursera.org/learn/financial-engineering-intro
  Term structure, credit and risk-neutral pricing (P vs Q).
- **MIT OpenCourseWare 18.S096: Mathematics with Applications in Finance** (Free full lecture videos)
  https://ocw.mit.edu/courses/18-s096-topics-in-mathematics-with-applications-in-finance-fall-2013/
  Lectures on portfolio theory, time series, credit and risk measures, taught by practitioners.

- **BIS: The Basel Framework** (Official rules (reading, not video))
  https://www.bis.org/basel_framework/
  The source for capital, leverage and liquidity ratios. Use it to check numbers, not to learn from scratch.

Bloomberg

- **Bloomberg for Education portal (Bloomberg Market Concepts)** (The actual course and tests)
  https://portal.bloombergforeducation.com/
  Sign in here, enter class code FM3NH456BD, and do the modules. The core BMC modules are Economic Indicators, Currencies, Fixed Income and Equities; check the course instructions to confirm these are your four required tests.
- **WallStreetMojo: Bloomberg Market Concepts beginner's guide** (Free written guide)
  https://wallstreetmojo.com/bloomberg-market-concepts-bmc
  Explains what each BMC module covers and how the tests work, before you start.
- **Khan Academy: Finance and capital markets** (Free short videos, very beginner)
  https://www.khanacademy.org/economics-finance-domain/core-finance
  Helps with the bond, stock and currency basics the BMC modules assume.
- **Yale (Coursera): Financial Markets with Robert Shiller** (Free to audit, video course)
  https://www.coursera.org/learn/financial-markets-global
  Background on how stock, bond and currency markets work and react to the economy.